

sized which fit the rhinovirus canyon and cause a change in conformation which prevents binding to cells and, since host proteins have very different folds to those of the viral capsid molecule, the drugs have limited cytotoxicity. A more recent drug has been designed to slot into the substrate-binding site of the neuraminidase, so inhibiting its biological activity.

Mutation can produce antagonistic T-cell epitopes

Hepatitis B virus isolates from chronically infected patients can present variant epitopes which act as TCR antagonists capable of inhibiting naturally occurring antiviral cytotoxic T-cells. Mutations which modify residues critical for recognition by MHC or TCR may generate partial agonists that can induce a profound and long-lasting state of T-cell anergy (cf. p. 233 and figure 9.8). Either strategy can lead to persistent infection.

Some viruses can affect antigen processing

Virtually every step in processing and presentation can be sabotaged by one virus or another. Thus, EBNA-1, the Epstein–Barr virus (EBV) nuclear antigen-1, contains glycine–alanine repeats which inhibit proteasome-mediated processing of the virus. Peptide binding to TAP is prevented by the infected cell protein 47 (ICP47) of herpes simplex virus, whilst the US6 protein of cytomegalovirus (CMV) prevents peptide transport through the TAP pore. Other CMV proteins can bind to and thereby retain MHC class I molecules in the endoplasmic reticulum; yet others redirect the molecules to degradation pathways. The K3 and K5 proteins of herpes virus downregulate expression of class I molecules. The net effect is a lack of surface target for cytotoxic T-cells. With respect to NK cells, which kill cells lacking MHC class I, CMV-encoded homologs of class I proteins can prevent NK-mediated lysis.

The MHC class II pathway is not exempt from such interference. HIV Nef affects vesicle traffic and endocytic processing involved in the generation of peptides, whilst the E1A protein of adenovirus interferes with IFN γ -mediated upregulation of MHC class II expression.

Viruses can interfere with immune effector mechanisms

Playing games with the host's humoral responses

Just as bacteria possess proteins capable of binding the Fc region of antibody (cf. p. 261), so certain viruses

also possess Fc γ receptors. Herpes simplex virus (HSV) types 1 and 2, coronavirus and murine cytomegalovirus all bear such molecules which, by binding antibody 'the wrong way round', may inhibit Fc-mediated effector functions.

As we saw for bacteria (cf. p. 255), viruses can block complement-mediated induction of the inflammatory response and thereby prevent viral killing. Pox viruses encode a complement control protein VCP (IMP) which binds C3b and C4b, making both the classical (C4b2a) and alternative (C3bBb) C3 convertases susceptible to factor I-mediated destruction. For its part, herpes simplex type 1 subverts the complement cascade by virtue of its surface glycoprotein C which binds C3b, interfering with its interaction with C5 and properdin.

Several viruses utilize complement receptors to gain entry into cells, especially since engagement of the complement receptor alone on a macrophage is a feeble activator of the respiratory burst. Flavivirus coated with iC3b enters through the CR3 receptors. Members of the regulators of the complement activation (RCA) family are used as cellular receptors for various viruses, such as CD46 (membrane cofactor protein) by measles virus and human herpes virus-6 (HHV-6), and CD55 (decay accelerating factor) used by echoviruses and coxsackie viruses. As noted previously, EBV infects B-cells by binding to the CR2 surface receptors. Ominously, HIV coated with antibody and complement is more virulent than unopsonized virus.

Cell-mediated immunity can also be manipulated

Parainfluenza virus type 2 strongly inhibits Tc cells by downregulating granzyme B expression (cf. p. 19). Numerous viral open reading frames encode proteins homologous to host cytokines and their receptors. Anti-IFN strategies are particularly abundant, with many viruses producing proteins able to block IFN-induced JAK/STAT pathway activation. A prime viral target is also the activation of the double-stranded RNA-dependent protein kinase (PKR) and other components of the cell thought to be involved in setting up an antiviral state in cells following their exposure to IFN. The macrophage tropic African swine fever virus encodes a homolog of I κ B and thereby blocks the NF κ B-mediated transcription of cytokine genes. Some viruses, including ectromelia (mousepox) virus, produce homologs of semaphorin molecules which bind to semaphorin receptors on macrophages, thereby inducing these signaling pathways in the host cells, a process which may facilitate viral replication. Human poxvirus-encoded proteins can bind IL-18 and thereby

inhibit IL-18-induced IFN γ production and NK responses. The list just goes on and on. Apoptosis of a cell could be considered bad news for a virus living very comfortably inside that cell. Therefore it is yet again not surprising that viruses have come up with ways of preventing apoptosis. Just a couple of examples. The adenovirus 14.7K gene encodes a protein which binds to and thereby inhibits caspase 8. Various herpesviruses encode homologs of the anti-apoptotic protein bcl-2. In contrast, some viral proteins may be pro-apoptotic, in this case perhaps aiding dissemination of virus particles.

An EBV product with 84% identity to human IL-10 helps the virus to escape the antiviral effects of IFN γ by downregulating Th1 cells. Poxviruses encode soluble homologs of the IFN α/β receptor and the IFN γ R, thereby competitively inhibiting the action of all three interferons. Others, particularly herpesviruses and poxviruses, possess several genes encoding chemokine-like and chemokine receptor-like proteins which can subvert the action of numerous chemokines.

Protection by serum antibody

The antibody molecule can neutralize viruses by a variety of means. It may stereochemically inhibit combination with the receptor site on cells, thereby preventing penetration and subsequent intracellular multiplication, the protective effect of antibodies to influenza viral hemagglutinin providing a good example. Similarly, antibodies to the measles hemagglutinin prevent entry into the cell, but the spread of virus from cell to cell is stopped by antibodies to the fusion antigen. Antibody may destroy a free virus particle directly through activation of the classical complement pathway or produce aggregation, enhanced phagocytosis and intracellular death by the mechanisms already discussed.

Relatively low concentrations of circulating antibody can be effective and one is familiar with the protection afforded by poliomyelitis antibodies, and by human γ -globulin given prophylactically to individuals exposed to measles. The most clear-cut protection is seen in diseases with long incubation times where the virus has to travel through the bloodstream before it reaches the tissue which it finally infects. For example, in poliomyelitis, the virus gains access to the body via the gastrointestinal tract and eventually passes through the circulation to reach the brain cells which become infected. Within the blood, the virus is neutralized by quite low levels of specific antibody, while the prolonged period before the virus infects the brain

allows time for a secondary immune response in a primed host.

Local factors

With other viral diseases, such as influenza and the common cold, there is a short incubation time, related to the fact that the final target organ for the virus is the same as the portal of entry and no intermediate stage involving passage through the body occurs. There is little time for a primary antibody response to be mounted and in all likelihood the **rapid production of interferon** is the most significant mechanism used to counter the viral infection. Experimental studies certainly indicate that, after an early peak of interferon production, there is a rapid fall in the titer of live virus in the lungs of mice infected with influenza (figure 13.19). Antibody, as assessed by the serum titer, seems to arrive on the scene much too late to be of value in aiding recovery. However, antibody levels may be elevated in the local fluids bathing the infected surfaces, e.g. nasal mucosa and lung, despite low serum titers, and it is the production of **antiviral antibody** (most prominently IgA) by locally deployed immunologically primed cells which is of major importance for the **prevention of subsequent infection**. Unfortunately, in so far as the common cold is concerned, a subsequent infection is likely to involve an antigenically unrelated virus so that general immunity to colds is difficult to achieve.

Cell-mediated immunity gets to the intracellular virus

In Chapter 2, we emphasized the general point that, to a first approximation, antibody deals with extracellu-

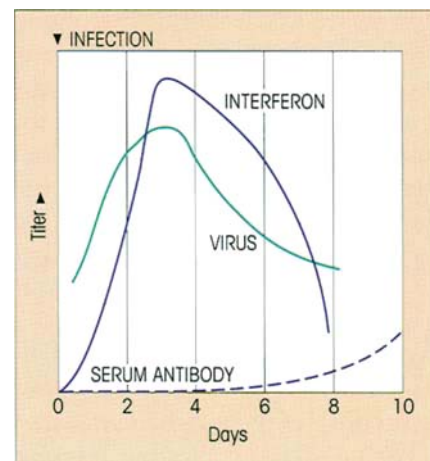


Figure 13.19. Appearance of interferon and serum antibody in relation to recovery from influenza virus infection of the lungs of mice. (From Isaacs A. (1961) *New Scientist* 11, 81.)

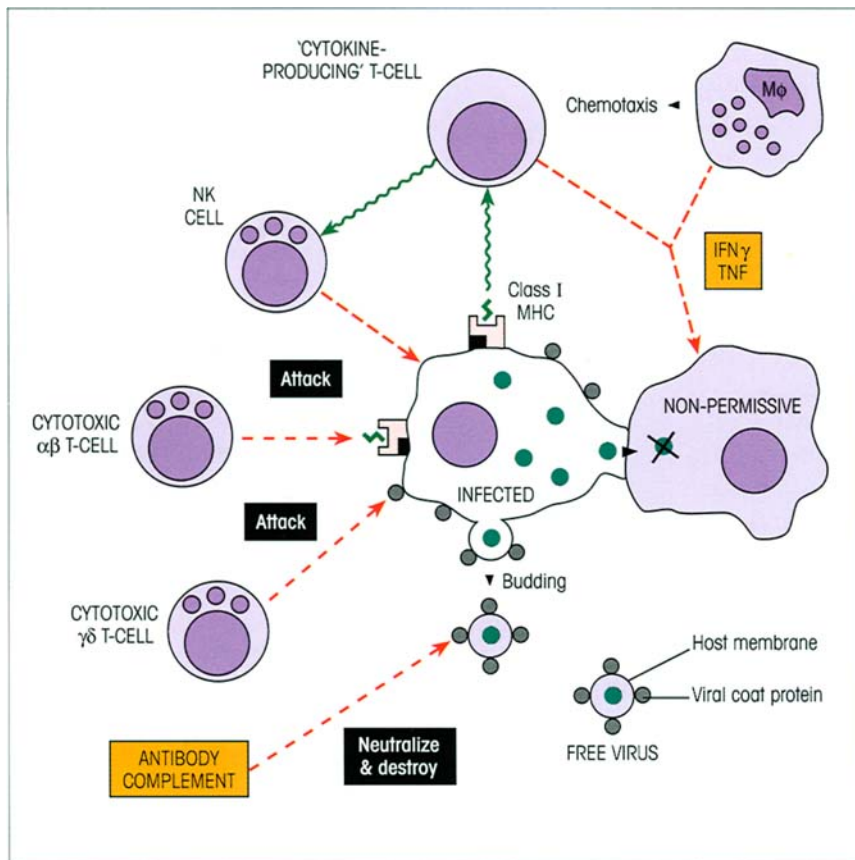


Figure 13.20. Control of infection by 'budding' viruses. Free virus released by budding from the cell surface is neutralized by antibody. Specific cytotoxic T-cells kill virally infected targets directly. Interaction with a (separate?) subpopulation of T-cells releases cytokines which attract macrophages, prime contiguous cells with IFN γ and TNF to make them resistant to viral infection and activate cytotoxic NK cells. NK cells are powerful producers of IFN γ and GM-CSF. They can recognize a lack of MHC class I on the infected cell membrane or partake in antibody-dependent cellular cytotoxicity (ADCC) if antibody to viral coat proteins is bound to the infected cell. Included in this group of budding viruses are: oncorna (=oncogenic RNA virus, e.g. murine leukemogenic), orthomyxo (influenza), paramyxo (mumps, measles), toga (dengue), rhabdo (rabies), arena (lymphocytic choriomeningitis), adeno, herpes (simplex, varicella zoster, cytomegalo, Epstein-Barr, Marek's disease), pox (vaccinia), papova (SV40, polyoma) and rubella viruses.

lar infective agents and CMI with intracellular ones. The same holds true for viruses which try to shelter from antibody in an intracellular habitat. Local or systemic antibodies can block the spread of cytolytic viruses which are released from the host cell they have just killed, but alone they are usually inadequate to control those viruses which bud off from the surface as infectious particles because they are also capable of spreading to adjacent cells without becoming exposed to antibody (figure 13.20). The importance of CMI for recovery from infection with these agents is underlined by the inability of children with primary T-cell immunodeficiency to cope with such viruses, whereas patients with Ig deficiency but intact CMI are not troubled in this way.

NK cells can kill virally infected targets

In earlier chapters, we have explained how early recognition and killing of a virally infected cell before replication occurs is of obvious benefit to the host. The importance of the NK cell in this role as an agent of pre-formed innate immunity can be gauged from observations on the exceedingly rare patients with complete absence of these cells who suffer recurrent life-

threatening viral infections, including EBV, varicella and cytomegaloviruses (CMVs). The surface of a virally infected cell undergoes modification, probably in its surface carbohydrate structures, making it an attractive target for NK cells, which can be shown to be cytotoxic *in vitro* to cells infected with a number of different viruses. The NK cell possesses two families of surface receptors. One, killer activating receptors, binds to carbohydrate and other structures expressed collectively by all cells; the other, killer inhibitory receptors, recognizes MHC class I molecules and overrules the signal from the activating receptor. Thus, the sensitivity of the target cell is closely related to self-MHC class I expression; sensitive targets have low class I, but transfection with the self-MHC molecule will generally protect them. Karrer has suggested that, whereas T-cells search for the presence of foreign shapes, NK cells survey tissues for the absence of self as indicated by aberrant or absent expression of MHC class I, which might occur in tumorigenesis or in certain viral infections. The ability of EBV to downregulate class I would explain the drastic nature of this infection in NK-deficient patients mentioned above. The production of IFN α during viral infection not only protects surrounding cells, but also activates NK cells

and upregulates MHC expression on the adjacent cells, making them more resistant to cytotoxicity.

Cytotoxic T-cells (Tc) are crucial elements in immunity to infection by budding viruses

T-lymphocytes from a sensitized host are directly cytotoxic to cells infected with viruses, the new MHC-associated peptide antigens on the target cell surface being recognized by specific $\alpha\beta$ receptors on the aggressor lymphocytes. There is a quite surprising frequency of dual specificities in target cell recognition by virus-specific Tc clones, which can lyse uninfected allogeneic cells or targets expressing peptides with little homology from different regions of the same viral protein, from different proteins of the same virus, or even from different unrelated viruses. Thus activation by a second virus may help to maintain memory, and there may be a spontaneous immunity to an unrelated virus after initial infection with a cross-reacting strain. Downregulation of MHC class I poses no problems for **TCR1 $\gamma\delta$ Tc** which recognize native viral coat protein (e.g. herpes simplex virus glycoprotein) on the cell surface (figure 13.20).

Tc cells can usually be detected in the peripheral blood lymphocytes of individuals who have recovered from infection with influenza, CMV or EBV by re-exposure *in vitro* to appropriately infected cells. In the case of CMV, for example, the targets are cells with the 'early antigen' on their surface expressed within 6 hours of infection. As discussed above, it is clearly advantageous for the cytotoxic cell to strike so soon after infection. Studies on volunteers, showing that high levels of cytotoxic activity before challenge with live influenza correlated with low or absent shedding of virus, speak in favor of the importance of Tc in human viral infection.

After a natural infection, both antibody and Tc cells are generated; subsequent protection is long-lived without reinfection, possibly being reinforced by bystander activation through cytokines released from other stimulated T-cells, or perhaps by random triggering with unrelated viruses based on the dual specificity described earlier. By contrast, injection of killed influenza produces antibodies but no Tc and protection is only short term.

Cytokines recruit effectors and provide a 'cordon sanitaire'

A number of studies on the transfer of protection to influenza, lymphocytic choriomeningitis, vaccinia, ectromelia and CMV infections have focused on CD8

rather than CD4 T-cells as the major defensive force. The knee-jerk response would be to implicate cytotoxicity, but remember that CD8 cells also produce cytokines. This may well be crucial when viruses escape the cytotoxic mechanism and manage to sidle laterally into an adjacent cell. CMI can now play some new cards: if T-cells stimulated by viral antigen release cytokines such as IFN γ and macrophage or monocyte chemokines, the mononuclear phagocytes attracted to the site will be activated to secrete TNF, which will synergize with the IFN γ to render the contiguous cells nonpermissive for the replication of any virus acquired by intercellular transfer (figure 13.20). In this way, the site of infection can be surrounded by a cordon of resistant cells. Like IFN α , IFN γ increases the nonspecific cytotoxicity of NK cells (see p. 18) for infected cells. This generation of 'immune interferon' (IFN γ) and TNF in response to non-nucleic acid viral components provides a valuable back-up mechanism when dealing with viruses which are intrinsically poor stimulators of interferon synthesis.

Antibody has a part too

The neutralization of free virus particles by antibody is relatively straightforward but the interaction with infected cells is rather more complex. Access to the surface antigens by $\alpha\beta$ T-cells cannot be blocked by antibody since these cells recognize processed antigen, whereas antibody binds native antigen. Antibodies can, however, block $\gamma\delta$ Tc by reacting with surface antigen on incipiently budding virions, but should be able to initiate ADCC (p. 32) as has been reported with herpes-, vaccinia- and mumps-infected target cells.

Do not forget the importance of antibody in **preventing reinfection** with most viruses.

IMMUNITY TO FUNGI

Not too much is known about this subject. Many fungal infections become established in immunocompromised hosts or when the normal commensal flora are upset by prolonged administration of broad-spectrum antibiotics. T-cells are important in defense and can recognize antigens from organisms such as *Cryptococcus neoformans* and *Candida albicans*, killing them by the granzyme system. NK cells can also lyse *C. neoformans*.

IMMUNITY TO PARASITIC INFECTIONS

The diverse organisms responsible for the major parasitic diseases are listed in figure 13.21. The numbers af-

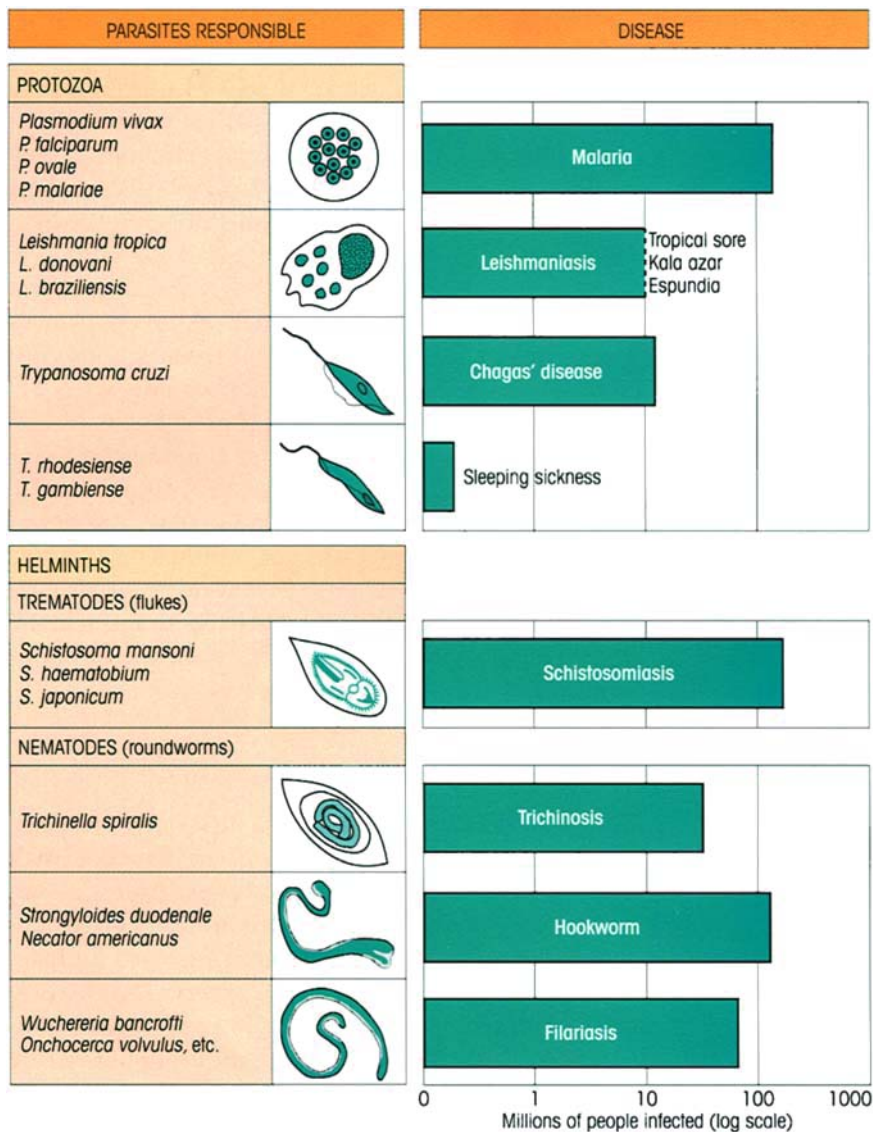


Figure 13.21. The major parasites in humans and the sheer enormity of the numbers of people infected. (Data from World Health Organization, 1990.)

infected are truly horrifying and the sum of misery these organisms engender is too large to comprehend. The consequences of parasitism could be, at one extreme, a lack of immune response leading to overwhelming superinfection, and, at the other, an exaggerated life-threatening immunopathologic response. To be successful, a parasite must steer a course *between* these extremes, avoiding wholesale killing of the human host and yet at the same time escaping destruction by the immune system. In practice, each type of parasite is virtually a world unto itself in the complexity of the mechanisms by which this is achieved.

The host responses

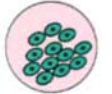
A wide variety of defensive mechanisms are deployed by the host, but the rough generalization may be made

that a humoral response develops when the organisms invade the bloodstream (malaria, trypanosomiasis), whereas parasites which grow within the tissues (e.g. cutaneous leishmaniasis) usually elicit CMI (table 13.1). Often, a chronically infected host will be resistant to reinfection with fresh organisms, a situation termed **concomitant immunity**. This is seen particularly in schistosomiasis but also in malaria, where historically the phenomenon was called 'premunity'. The resident and the infective forms must differ in some way yet to be pinpointed.

Humoral immunity

Antibodies of the right specificity present in adequate concentrations and affinity are reasonably effective in providing protection against blood-borne parasites, such as *Trypanosoma brucei*, and the sporozoite and

Table 13.1. The relative importance of antibody and cell-mediated responses in protozoal infections.

PARASITE	TRYPANOSOMA BRUCEI	PLASMODIUM	TRYPANOSOMA CRUZI	LEISHMANIA
HABITAT	Free in blood 	Inside red cell 	Inside macrophage 	Inside macrophage 
ANTIBODY				
Importance	++++	+++	++	+
Mechanism	Lysis with complement Opsonizes for phagocytosis	Blocks invasion Opsonizes for phagocytosis	Limits spread in acute infection	Limits spread
Means of evasion	Antigenic variation	Intracellular habitat Antigenic variation	Intracellular habitat	Intracellular habitat
CELL-MEDIATED				
Importance	—	+	+++ (Chronic phase)	++++
Mechanism	—	Cytokine-mediated activation of macrophages and NK cells	Macrophage activation by cytokines and killing by TNF, metabolites of O ₂ and NO- Role for cytotoxic T-cells	

merozoite stages of malaria. Thus, individuals receiving IgG from solidly immune adults in malaria endemic areas are themselves temporarily protected against infection, the effector mechanisms being opsonization and phagocytosis, and complement-dependent lysis.

A marked feature of the immune reaction to helminthic infections, such as *Trichinella spiralis* in humans and *Nippostrongylus brasiliensis* in the rat, is the eosinophilia and the high level of IgE antibody produced. In humans, serum levels of IgE can rise from normal values of around 100 ng/ml to as high as 10 000 ng/ml. These changes have all the hallmarks of response to Th2-type cytokines (cf. p. 181) and it is notable that, in animals infected with helminths, injection of anti-IL-4 greatly reduces IgE production and anti-IL-5 suppresses the eosinophilia. This exceptional increase in IgE has encouraged the view that it represents an important line of defense. One can see that antigen-specific triggering of IgE-coated mast cells would lead to exudation of serum proteins containing high concentrations of protective antibodies in all the major Ig classes and the release of eosinophil chemotactic factor. It is relevant to note that schistosomula, the early immature form of the schistosome, have been killed in cultures containing both specific IgG and eosinophils, which induce a form of ADCC by binding through their FcγRII receptors to the IgG-coated organism (figure 13.22); after 12 hours or so, the major basic protein forming the electron-dense core of

the eosinophilic granules is released onto the parasite and brings about its destruction. A contribution to this process from CMI is emerging, since eosinophils can express class II MHC and their IgG-mediated ADCC is strongly enhanced by granulocyte-macrophage colony-stimulating factor (GM-CSF) and TNF. Further evidence for an involvement of this cell comes from the experiment in which the protection afforded by passive transfer of antiserum *in vivo* was blocked by pretreatment of the recipient with an anti-eosinophil serum. It has also been found that eosinophils can kill IgE-coated schistosomula, but the mechanism is different because activation of the IgE (FcεRI) receptors now triggers release of platelet activating factors and the eosinophil peroxidase. This dichotomy in Fcγ and Fcε receptor pathways is also evident from reports that IgE but not IgG can mediate schistosome killing by macrophages or platelets.

Two further points are in order. The IgE-mediated reactions may be vital for recovery from infection, whereas the resistance in vaccinated hosts may be more dependent upon preformed IgG and IgA antibodies.

Cell-mediated immunity

Just like mycobacteria, many parasites have adapted to life within the macrophage despite the possession by that cell of potent microbicidal mechanisms includ-

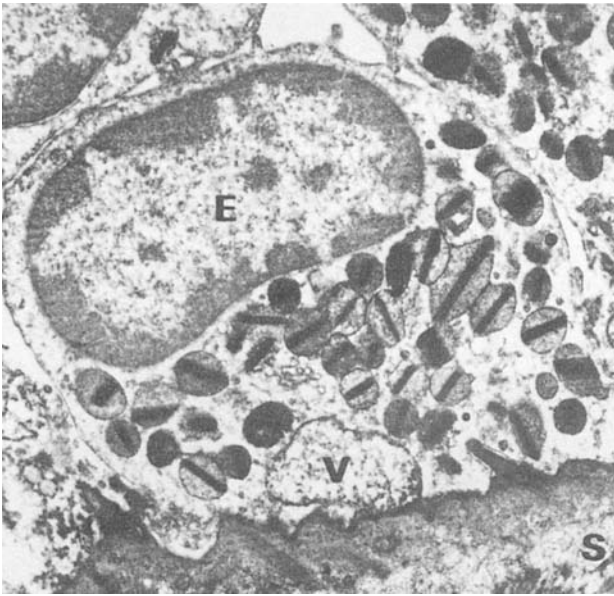


Figure 13.22. Electron micrograph showing an eosinophil (E) attached to the surface of a schistosomulum (S) in the presence of specific antibody. The cell develops large vacuoles (V) which appear to release their contents on to the parasite ($\times 16500$). (Courtesy of Drs D.J. McLaren and C.D. Mackenzie.)

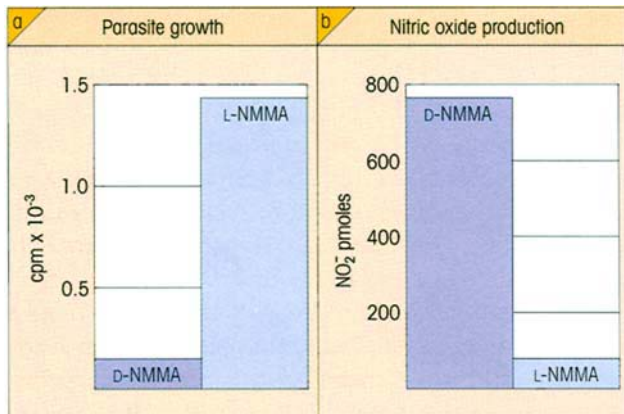


Figure 13.23. Role of $\text{NO}\cdot$ in macrophage leishmanicidal activity. The $\text{NO}\cdot$ synthase inhibitor L-NMMA (50 mM) inhibits the ability of macrophages to kill intracellular *Leishmania* where growth is monitored by [^3H]thymidine incorporation (a) and also blocks NO production measured by accumulation of NO_2^- in the culture supernatant at 72 hours (b). The D-isomer, D-NMMA, used as a control does not inhibit the enzyme. (Data taken from Liew F.Y. & Cox F.E.G. (1991) *Immunology Today* 12, A17.)

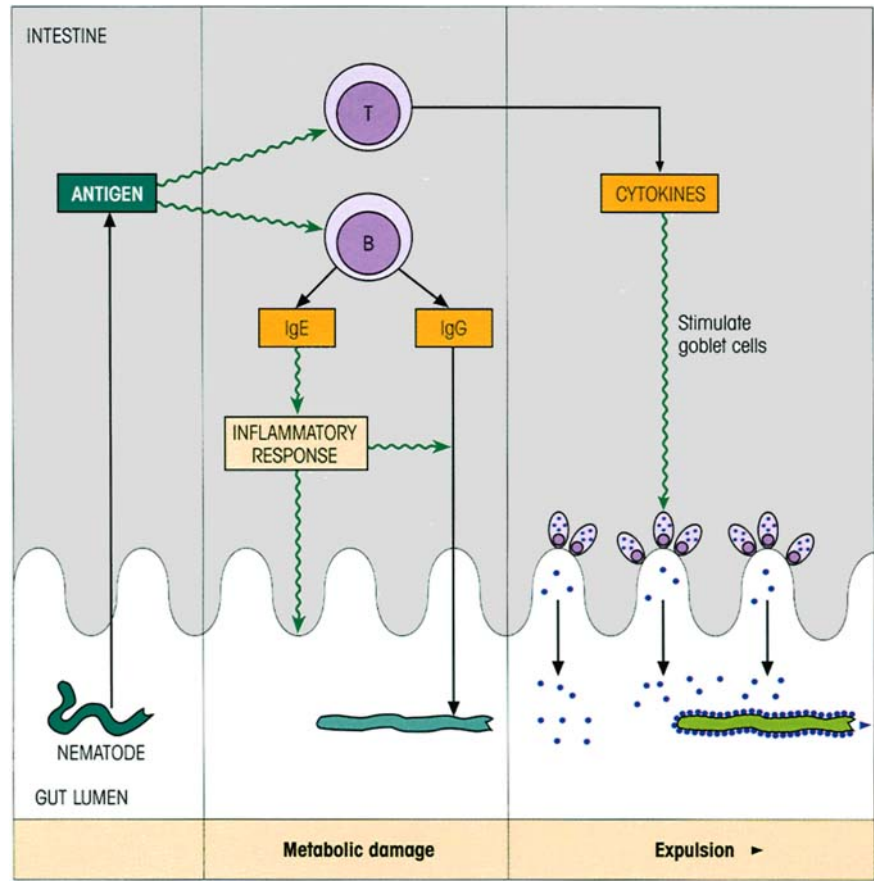
ing $\text{NO}\cdot$ (figure 13.23). Intracellular organisms, such as *Toxoplasma gondii*, *Trypanosoma cruzi* and *Leishmania* spp., use a variety of ploys to subvert the macrophage killing systems (see below) but again, as with mycobacterial infections, cytokine-producing T-cells are crucially important for the stimulation of macrophages to release their killing power and dispose of the unwanted intruders.

In vivo, the balance of cytokines produced is of the utmost importance. Infection of mice with *Leishmania major* is instructive in this respect; the organism produces fatal disease in susceptible mice but other strains are resistant. This is partly controlled by alleles of the *Nramp-1* gene (cf. p. 266) but, as discussed earlier in Chapter 10, in susceptible mice there is excessive stimulation of Th2 cells producing IL-4 which do not help to eliminate the infection, whereas resistant strains are characterized by the expansion of Th1 cells which secrete $\text{IFN}\gamma$ in response to antigen presented by macrophages harboring *living* protozoa. Combined therapy of susceptible strains with the leishmanicidal drug, Pentostam, plus IL-12, which recruits Th1 cells, provides promise that Th2 activities which exacerbate disease can be switched to protective Th1 responses. CD4 clones which recognize only *lysates* of the organism do not confer protection even though they produce $\text{IFN}\gamma$, a point to be borne in mind in designing vaccines. Experiments in knockout mice suggest that perforins, granzyme B and FasL are not necessary for the protection afforded to mice by vaccination with irradiated sporozoites. Of particular importance for protection, however, is the induction of $\text{IFN}\gamma$ and CD8^+ T-cells. Interleukin-12 and nitric oxide are also required, and NK cells may play a subsidiary role.

Organisms such as malarial plasmodia, and incidentally rickettsiae and chlamydiae, that live in cells which are not professional phagocytes, may be eliminated through activation of intracellular defense mechanisms by $\text{IFN}\gamma$ released from CD8^+ T-cells or even by direct cytotoxicity. This is very much the case in hepatic cells harboring malarial sporozoites, and it is pertinent to note that, following the recognition of an association between HLA-B53 and protection against severe malaria, B53-restricted Tc reacting with a conserved nonamer from a liver stage-specific antigen were demonstrated in the peripheral blood of resistant individuals. A large case control study of malaria in Gambian children showed that the protective B53 class I antigen is common in West African children but rare in other racial groups, lending further credence to the hypothesis that MHC polymorphism has evolved primarily through natural selection by infectious pathogens.

Eliminating worm infestations of the gut is a more tricky operation and the combined forces of cellular and humoral immunity are required to expel the unwanted guest. One of the models studied is the response to *Nippostrongylus brasiliensis*; transfer studies in rats showed that, although antibody produces some damage to the worms, T-cells from *immune* donors are also required for vigorous expulsion, which is proba-

Figure 13.24. The expulsion of nematode worms from the gut. The parasite is first damaged by IgG antibody passing into the gut lumen, perhaps as a consequence of IgE-mediated inflammation and possibly aided by accessory ADCC cells. Cytokines released by antigen-specific triggering of T-cells stimulate proliferation of goblet cells and secretion of mucous materials, which coat the damaged worm and facilitate its expulsion from the body by increased gut motility induced by mast cell mediators, such as leukotriene-D₄, and diarrhea resulting from inhibition of glucose-dependent sodium absorption by mast cell-derived histamine and PGE₂.



bly achieved through a combination of mast cell-mediated stimulation of intestinal motility and cytokine activation of the innumerable intestinal goblet cells. These secrete a complex mixture of densely glycosylated high molecular weight molecules which form a viscoelastic gel around the worm, so protecting the colonic and intestinal surfaces from invasion (figure 13.24). Another model, this time of *Trichinella spiralis* infection in mice, again hints at a duality of T-subset cytokine responses. One strain, which expels adult worms rapidly, makes large amounts of IFN γ and IgG2a antibody, while, in contrast, more susceptible mice make miserly amounts of IFN γ and favor IgG1, IgA and IgE antibody classes. Clearly, the protective strategy varies with the infection.

Evasive strategies by the parasite

Resistance to effector mechanisms

Some tricks to pre-empt the complement defenses are of interest. *T. cruzi* has elegantly created a DAF-like molecule (cf. p. 307) which accelerates the decay of C3b. The cercariae of *Schistosoma mansoni* activate com-

plement directly, but eject the bound C3 by shedding their glycocalyx. The *Plasmodium falciparum* protein PfEMP1 is expressed on the surface of infected erythrocytes and can bind to CR1 (CD35) on other infected erythrocytes leading to rosette formation, which may facilitate spread of the parasite with minimal exposure to the host immune system. In a similar fashion, malarial sporozoites shed their circumsporozoite antigen when it binds antibody, and *Trypanosoma brucei* releases its surface antigens into solution to act as decoy proteins (p. 255). In each case, these shedding and decoy systems are well suited to parasites or stages in the parasite life cycle which are only briefly in contact with the immune system.

We have already mentioned the way in which different protozoal parasites hide away from the effects of antibody by using the interior of a macrophage as a sanctuary. To do this they must block the normal microbicidal mechanisms and they use similar methods to those deployed by intracellular obligate and facultative bacteria (cf. p. 263). *Toxoplasma gondii* inhibits phagosome-lysosome fusion by lining up host cell mitochondria along the phagosome membrane. *Trypanosoma cruzi* escapes from the confines of the

phagosome into the cytoplasm, while *Leishmania* parasites are surrounded by a lipophosphoglycan which protects them from the oxidative burst by scavenging oxygen radicals. They also downregulate expression of MHC and B7 so diminishing T-cell stimulation.

Avoiding antigen recognition by the host

Some parasites **disguise** themselves to look like the host. This can be achieved by molecular mimicry as demonstrated by cross-reactivity between *Ascaris* antigens and human collagen. Another way is to cover the surface with host protein. Schistosomes are very good at that; the adult worm takes up host red-cell glycoproteins, MHC molecules and IgG and lives happily in the mesenteric vessels of the host, despite the fact that the blood which bathes it contains antibodies which can prevent reinfection.

Another very crafty ruse, rather akin to moving the goalposts in football, is **antigenic variation**, in which the parasites escape from the cytotoxic action of humoral antibody on their cycling blood forms by the ingenious trick of altering their antigenic constitution. Figure 13.25 illustrates how the trypanosome continues to infect the host, even after fully protective antibodies appear, by switching to the expression of a new antigenic variant which these antibodies cannot inactivate; as antibodies to the new antigens are synthesized, the parasite escapes again by changing to yet a further variant and so on. In this way, the parasite can remain in the bloodstream long enough to allow an opportunity for transmission by blood-sucking insects or blood-

to-blood contact. The same phenomenon has been observed with *Plasmodium* spp. and this may explain why, in hyperendemic areas, children are subjected to repeated attacks of malaria for their first few years and are then solidly immune to further infection. Immunity must presumably be developed against all the antigenic variants before full protection can be attained, and indeed it is known that IgG from individuals with solid immunity can effectively terminate malaria infections in young children.

Deviation of the host immune response

Immunosuppression has been found in most of the parasite infections studied. During infection by trypanosomes, for example, antibody and CMI are only 5–10% of the normal values while T-suppressor activity mediated by IL-10 and IFN γ is prominent, presumably related to an excessive load of antigen. *Schistosoma mansoni* possesses a gene with homology for proopiomelanocortin which, in the human, is a prohormone that is cleaved to generate adrenocorticotrophic hormone (ACTH), melanocyte-stimulating hormone (MSH) and the opioid β -endorphin. All have immunomodulatory properties and both ACTH and β -endorphin have been demonstrated in culture when adult worms were incubated at 37°C in minimum essential medium; if neutrophils are added to the system, α -MSH, derived from ACTH by the cell's neutral endopeptidase (CD10; CALLA, p. 387), is also formed.

Parasites may also manipulate T-cell subsets to their own advantage. Filariasis provides a case in point: it

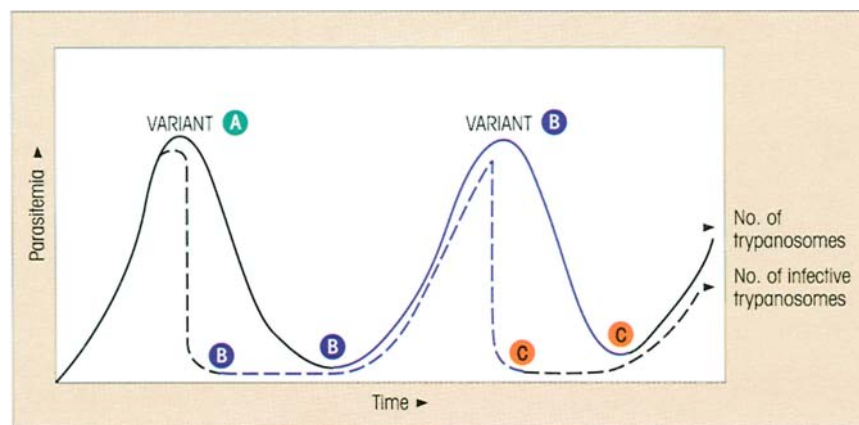


Figure 13.25. Antigenic variation during chronic trypanosome infection. As antibody to the initial variant A is formed, the blood trypanosomes become complexed prior to phagocytosis and are no longer infective, leaving a small number of viable parasites which have acquired a new antigenic constitution. This new variant (B) now multiplies until it, too, is neutralized by the primary antibody

response and is succeeded by variant C. At any time, only one of the variant surface glycoproteins (VSGs) is expressed and covers the surface of the protozoan to the exclusion of all other antigens. Nearly 9% of the genome (approximately 1000 genes) is devoted to generation of VSGs. Switching occurs by insertion of a duplicate gene into a new genomic location in proximity to the promoter.

has been suggested that individuals with persistent microfilariae fail to mount presumably protective immediate hypersensitivity responses, including IgE and eosinophilia, as a result of active suppression of Th2 cells.

Epidemiological surveys accord with a protective role for IgE antibodies in schistosomiasis, but they also reveal a susceptible population producing IgM and IgG4 antibodies which can block ADCC dependent upon IgE. The ability of certain helminths to activate IgE-producing B-cells polyclonally is good for the parasite and correspondingly not so good for the host, since a high concentration of irrelevant IgE binding to a mast cell will crowd out the parasite-specific IgE molecules and diminish the possibility of triggering the mast cell by specific antigen to initiate a protective defensive reaction.

Transmissible spongiform encephalopathies

New variant Creutzfeldt–Jakob disease (nvCJD) was first described in 1996 and, in common with sheep scrapie and bovine spongiform encephalopathy (BSE), is classed as a transmissible spongiform encephalopathy (TSE) caused by prions. The role of the immune system in prion diseases seems to be one of helping the disease rather than combating it. These infectious diseases lead to abnormally folded, relatively protease-resistant forms of host prion protein (PrP). Infectivity usually replicates to high levels in lymphoid tissues before spreading to the central nervous system, and there is evidence to suggest that follicular dendritic cells (FDCs) in spleen, lymph node and Peyer's patches may be involved in this replication. This may be because FDCs naturally express high levels of the normal PrP which then becomes abnormal following exposure

to the TSE agent. B-lymphocytes may play a subsidiary role via their production of lymphotoxin, a cytokine necessary for the maintenance of a differentiated state in the FDCs. Furthermore, in addition to FDCs, macrophages have been proposed as providing a reservoir of infectivity.

Immunopathology

Where parasites persist chronically in the face of an immune response, the interaction with foreign antigen frequently produces tissue-damaging reactions. One example is the immune complex-induced nephrotic syndrome of Nigerian children associated with quartan malaria. Increased levels of TNF are responsible for pulmonary changes in acute malaria, cerebral malaria in mice and severe wasting of cattle with trypanosomiasis. Another example is the liver damage resulting from IL-4-mediated granuloma formation around schistosome eggs (cf. figure 16.28); one of the egg antigens directly induces IL-10 production in B-cells, thereby contributing to Th2 dominance. Remarkably, the hypersensitivity reaction helps the eggs to escape from the intestinal blood capillaries into the gut lumen to continue the cycle outside the body, an effect mediated by TNF α .

Cross-reaction between parasite and self may give rise to autoimmunity, and this has been proposed as the basis for the cardiomyopathy in Chagas' disease. It is also pertinent that the nonspecific immunosuppression which is so widespread in parasitic diseases tends to increase susceptibility to bacterial and viral infections and, in this context, the association between Burkitt's lymphoma and malaria has been ascribed to an inadequate host response to the Epstein–Barr virus.

SUMMARY

Immunity to infection involves a constant battle between the host defenses and the mutant microbes trying to evolve evasive strategies.

Inflammation revisited

- Inflammation is a major defensive reaction initiated by infection or tissue injury.
- The mediators released upregulate adhesion molecules on endothelial cells and leukocytes, which pair together causing, first, rolling of leukocytes along the vessel wall

and then passage across the blood vessel up the chemotactic gradient to the site of inflammation.

- IL-1, TNF and chemokines such as IL-8 are involved in maintaining the inflammatory process.
- Inflammation is controlled by complement regulatory proteins, PGE₂, TGF β , glucocorticoids and IL-10.
- LPS is bound by LBP which transfers the LPS to the CD14–TLR4 complex, thereby activating genes in the APC which encode proinflammatory molecules.
- Inability to eliminate the initiating agent leads to a

(continued p. 278)

chronic inflammatory response dominated by macrophages often forming granulomas.

Extracellular bacteria susceptible to killing by phagocytosis and complement

- Bacteria try to avoid phagocytosis by surrounding themselves with capsules, secreting exotoxins which kill phagocytes or impede inflammatory reactions, deviating complement to inoffensive sites or by colonizing relatively inaccessible locations.
- Antibody combats these tricks by neutralizing the toxins, making complement deposition more even on the bacterial surface, and overcoming the antiphagocytic nature of the capsules by opsonizing them with Ig and C3b.
- The secretory immune system protects the external mucosal surfaces. IgA inhibits adherence of bacteria and can opsonize them. IgE bound to mast cells can initiate the influx of protective IgG, complement and polymorphs to the site by a miniature acute inflammatory response.

Bacteria which grow in an intracellular habitat

- Intracellular bacteria such as tubercle and leprosy bacilli grow within macrophages. They defy killing mechanisms by blocking macrophage activation, scavenging oxygen radicals, inhibiting lysosome fusion, having strong outer coats and by escaping from the phagosome into the cytoplasm.
- They are killed by CMI: specifically sensitized T-helpers release cytokines on contact with infected macrophages which powerfully activate the formation of nitric oxide (NO $\cdot$), reactive oxygen intermediates (ROIs) and other microbicidal mechanisms.

Immunity to viral infection

- Viruses try to avoid the immune system by changes in the antigenicity of their surface antigens. Point mutations bring about minor changes (antigenic drift), but radical changes leading to endemics can result from wholesale swapping of genetic material with different viruses in other animal hosts (antigenic shift).
- Some viruses subvert the function of the complement system to their own advantage.
- Antibody neutralizes free virus and is particularly effective when the virus has to travel through the bloodstream before reaching its final target.
- Where the target is the same as the portal of entry, e.g. the lungs, IFN is dominant in recovery from infection.
- Antibody is important in preventing reinfection.
- 'Budding' viruses which can invade lateral cells without becoming exposed to antibody are combated by CMI.

Infected cells express a processed viral antigen peptide on their surface in association with MHC class I a short time after entry of the virus, and rapid killing of the cell by cytotoxic $\alpha\beta$ T-cells prevents viral multiplication which depends upon the replicative machinery of the intact host cell. $\gamma\delta$ Tc recognize native viral coat protein on the target cell surface. NK cells are also cytotoxic.

- T-cells and macrophages producing IFN γ and TNF bathe the contiguous cells and prevent them from becoming infected by lateral spread of virus.

Immunity to parasitic infections

- Diseases involving *protozoal parasites* and *helminths* affect hundreds of millions of people. Antibodies are usually effective against the blood-borne forms. IgE production is notoriously increased in worm infestations and can lead to mast cell-mediated influx of Ig and eosinophils; schistosomes coated with IgG or IgE are killed by adherent eosinophils through extracellular mechanisms involving the release of cationic proteins and peroxidase.
- Organisms such as *Leishmania* spp., *Trypanosoma cruzi* and *Toxoplasma gondii* hide from antibodies inside macrophages, use the same strategies as intracellular parasitic bacteria to survive, and like them are killed when the macrophages are activated by Th1 cytokines produced during cell-mediated immune responses. NO $\cdot$ is an important killing agent.
- CD8 cells also have a protective role.
- Expulsion of intestinal worms usually depends heavily on Th2 responses and requires the coordinated action of antibody, the release of mucin by cytokine-stimulated goblet cells and the production of intestinal contraction and diarrhea by mast cell mediators.
- Some parasites avoid recognition by disguising themselves as the host, either through molecular mimicry or by absorbing host proteins to their surface.
- Other organisms such as *Trypanosoma brucei* and various malarial species have the extraordinary ability to cover their surface with a dominant antigen which is changed by genetic switch mechanisms to a different molecule as antibody is formed to the first variant.
- Most parasites also tend to produce nonspecific suppression of host responses.
- Chronic persistence of parasite antigen in the face of an immune response often produces tissue-damaging immunopathological reactions such as immune complex nephrotic syndrome, liver granulomas and autoimmune lesions of the heart. Generalized immunosuppression increases susceptibility to bacterial and viral infections.

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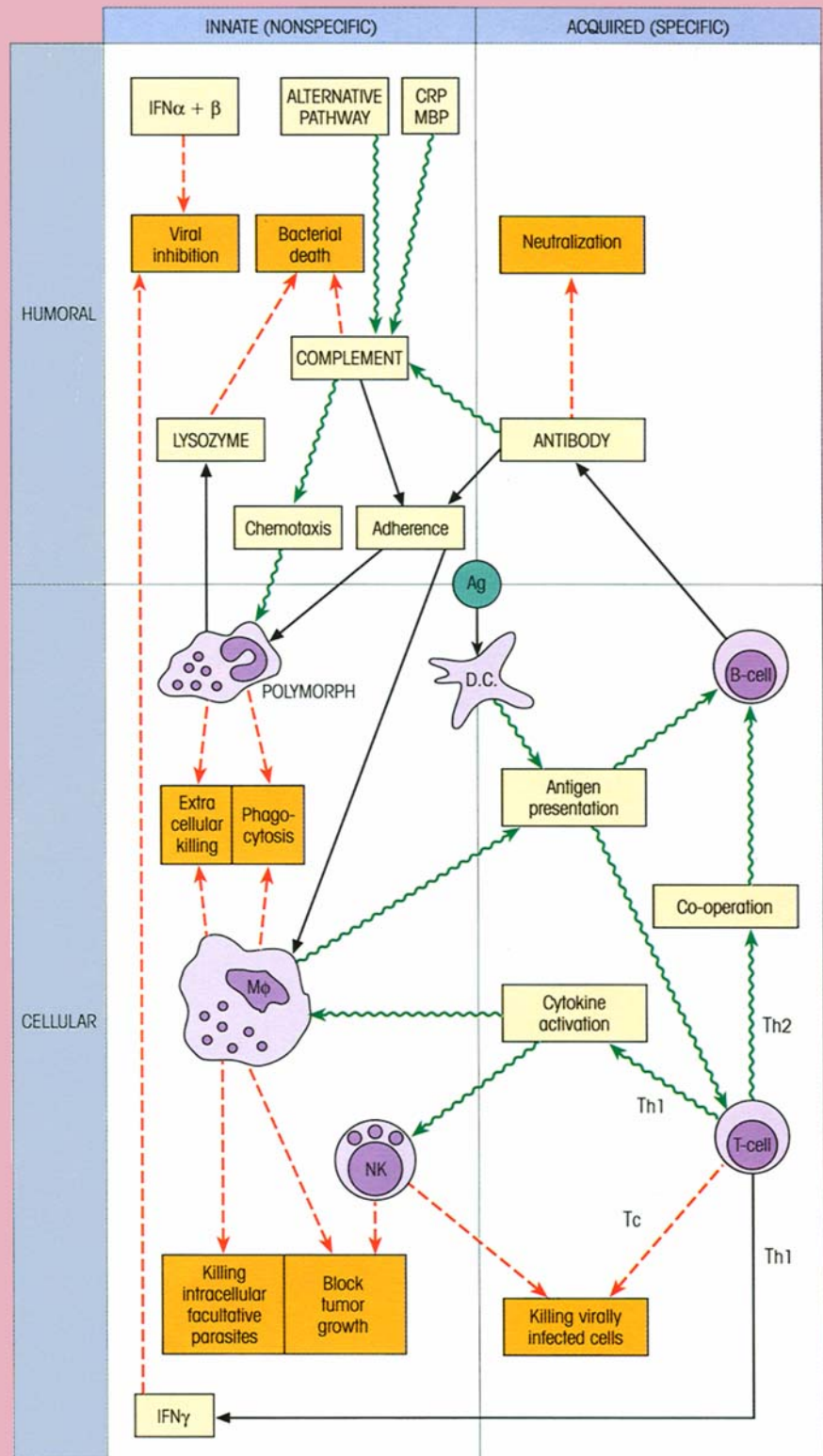


Figure 13.26. Simplified scheme to emphasize the interactions between innate and acquired immunity mechanisms. The dendritic cell which presents antigen to B-cells in the form of immune complexes is the follicular dendritic cell in germinal centers, whereas the MHC class II-positive interdigitating dendritic cell presents antigen to T-cells. (Developed from Playfair J.H.L. (1974) *British Medical Bulletin* 30, 24.)

(continued p. 280)

- As the features of the response to infection are analysed, we see more clearly how the specific acquired response operates to amplify and enhance innate immune mechanisms; the interactions are summarized in figure 13.26.

Prion diseases

- Scrapie, BSE and nvCJD are transmissible spongiform encephalopathies caused by prions.

- Abnormally folded, protease-resistant forms of host prion protein (PrP) develop.
- FDCs in lymphoid tissues become infected prior to spread of the infectious agent to the CNS.

See the accompanying website (www.roitt.com) for multiple choice questions.

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INTRODUCTION

The control of infection is approached from several directions. One method of breaking the chain of infection has been achieved in the UK with rabies and psittacosis by controlling the importation of dogs and parrots, respectively. Improvements in public health—water supply, sewage systems, education in personal hygiene—prevent the spread of cholera and many other diseases; and of course when other measures fail we can fall back on the induction of immunity (Milestone 14.1).

PASSIVELY ACQUIRED IMMUNITY

Temporary protection against infection can be established by giving preformed antibody from another individual of the same or a different species (table 14.1; figure 14.1). As the acquired antibodies are utilized by combination with antigen or catabolized in the normal way, this protection is gradually lost.

Horse globulins containing antitetanus and antidiphtheria toxins have been extensively employed prophylactically, but at the present time the practice is

more restricted because of the complication of serum sickness developing in response to the foreign protein. This is more likely to occur in subjects already sensitized by previous contact with horse globulin; thus individuals who have been given horse antitetanus (e.g. for immediate protection after receiving a wound out in the open) are later advised to undergo a course of active immunization to obviate the need for further injections of horse protein in any subsequent emergency.

Maternally acquired antibody

In the first few months of life, while the baby's own lymphoid system is slowly getting under way, protection is afforded to the fetus by maternally derived IgG antibodies acquired by placental transfer and to the neonate by intestinal absorption of colostral immunoglobulins (figure 14.1). The major immunoglobulin in milk is secretory IgA and this is not absorbed by the baby but remains in the intestine to protect the mucosal surfaces. In this respect it is quite striking that the sIgA antibodies are directed against bacterial and viral antigens often present in the intestine, and it is presumed that IgA-producing cells, responding to gut

Milestone 14.1—Vaccination

The notion that survivors of serious infectious disease seldom contract that infection again has been embedded in folklore for centuries. In an account of the terrible plague which afflicted Athens, Thucydides noted that, in the main, those nursing the sick were individuals who had already been infected and yet recovered from the plague. Deliberate attempts to ward off infections by inducing a minor form of the disease in otherwise healthy subjects were common in China in the Middle Ages. There, they developed the practice of inhaling a powder made from **smallpox** scabs as protection against any future infection. The Indians inoculated the scab material into small skin wounds, and this practice of **variolation** (Latin *varus*, a pustular facial disease) was introduced into Turkey where the inhabitants of Circassia were determined to prevent the ravages of smallpox epidemics interfering with the lucrative sale of their gorgeous daughters to the harems of the wealthy.

Voltaire, in 1773, tells us that the credit for spreading the practice of variolation to Western Europe should be attributed to Lady Wortley Montague, a remarkably enterprising woman who was the wife of the English Ambassador to Constantinople in the time of George I. With little scruple, she inoculated her daughter with smallpox in the face of the protestations of her Chaplain who felt that it could only succeed with infidels, not Christians. All went well however and the practice was taken up in England despite the hazardous nature of the procedure which had a case fatality of 0.5–2%. These dreadful risks were taken because, at that time, as Voltaire recorded ‘... three score persons in every hundred have the smallpox. Of these three score, twenty die of it in the most favorable season of life, and as many more wear the disagreeable remains of it on their faces so long as they live.’

Edward Jenner (1749–1823), a country physician in Gloucestershire, suggested to one of his patients that she might have smallpox, but she assured him that his diagnosis was impossible since she had already contracted cowpox through her chores as a milkmaid (folklore again!). This led Jenner to the series of experiments in which he showed that prior inoculation with cowpox, which was nonvirulent (i.e. nonpathogenic) in the human, protected against subsequent challenge with smallpox (cf. p. 30). His ideas initially met with violent opposition but were even-



Figure M14.1.1. Edward Jenner among patients in the Smallpox and Inoculation Hospital at St Pancras. Etching after J. Gillray, 1802. (Kindly supplied by The Wellcome Centre Medical Photographic Library, London.)

tually accepted and he achieved world fame; learned societies everywhere elected him to membership, although it is intriguing to note that the College of Physicians in London required him to pass an examination in classics and the Royal Society honored him with a Fellowship on the basis of his work on the nesting behavior of the cuckoo. In the end he inoculated thousands in the shed in the garden of his house in Berkeley, Gloucestershire, which now functions as a museum and venue for small symposia organized by the British Society for Immunology (rather fun to visit if you get the chance).

The next seminal development in vaccines came through the research of Louis Pasteur who had developed the germ theory of disease. A culture of chicken cholera bacillus, which had accidentally been left on a bench during the warm summer months, lost much of its ability to cause disease; nonetheless, birds which had been inoculated with this old culture were resistant to fresh virulent cultures of the bacillus. This **attenuation** of **virulent** organisms was reproduced by Pasteur for anthrax and rabies using abnormal culture and passage conditions. Recognizing the relevance of Jenner's research for his own experiments, Pasteur called his treatment **vaccination**, a term which has stood the test of time.

Figure 14.1. Passive immunization produced by: transplacental passage of IgG from mother to fetus, acquisition of IgA from mother's colostrum and milk by the infant, and injection of polyclonal antibodies, recombinant monoclonal antibodies expressed in prokaryotes or plants, antibody fragments (Fab or scFv) derived from phage libraries and, in the future, synthetic polymorph defensins.

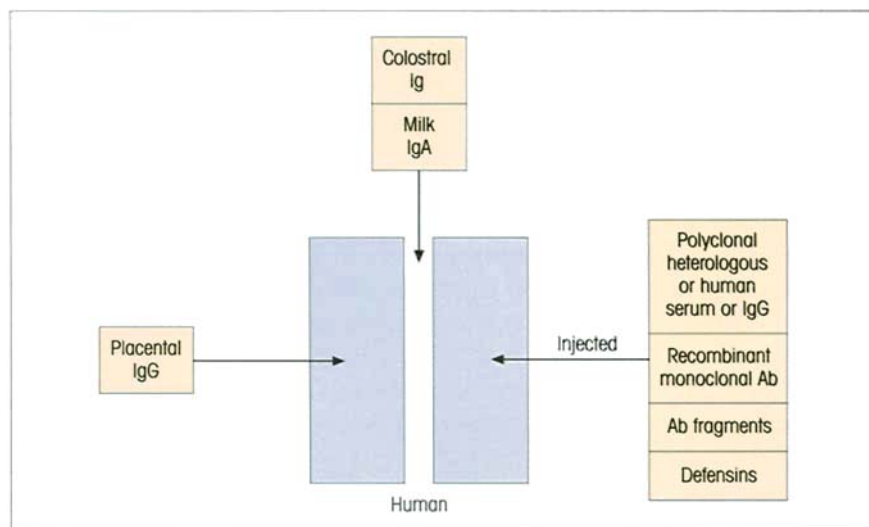


Table 14.1. Passive immunotherapy with antibody.

INFECTION	SOURCE OF ANTIBODY		USE
	HORSE	HUMAN	
Tetanus Diphtheria	✓	✓	Prophylaxis Treatment
Botulism Gas gangrene Snake or scorpion bite	✓	—	Treatment
Varicella zoster	—	✓	Treatment immunodeficiency
Rabies	—	✓	Post-exposure to vaccine
Hepatitis B	—	✓	Treatment
Hepatitis A	—	✓	Prophylaxis (Travel)
Measles	—	✓	Treatment
Cytomegalovirus	—	✓	Prophylaxis in patients receiving immunosuppression

antigens, migrate and colonize breast tissue (as part of the MALT immune system; see p. 154), where the antibodies they produce appear in the milk. The case for mucosal vaccination of future mothers against selected infections is inescapable.

Pooled human γ -globulin

Regular injection of pooled human adult γ -globulin is an essential treatment for patients with long-standing immunodeficiency. The preparations are also of value to modify the effects of chickenpox or measles in other individuals with defective immune responses, such as premature infants, children with protein malnutrition or patients on steroid treatment. Contacts with

cases of infectious hepatitis may also be afforded protection by γ -globulin. Human antitetanus immunoglobulin is preferable to horse antitoxin serum, which may cause hypersensitivity reactions. Curiously, pooled γ -globulin is being increasingly used as a treatment for autoimmune diseases such as idiopathic thrombocytopenic purpura, possibly acting through anti-idiotypic mechanisms.

Isolated γ -globulin preparations tend to form small aggregates spontaneously and these can lead to severe anaphylactic reactions when administered intravenously, on account of their ability to aggregate platelets and to activate complement and generate C3a and C5a anaphylatoxins. For this reason, the material is always injected intramuscularly. Preparations free of aggregates are available, and separate pools with raised antibody titers to selected organisms such as vaccinia, herpes zoster, tetanus and perhaps rubella would be welcome. This need may ultimately be satisfied as it becomes possible to produce human monoclonal antibodies on demand.

Cultured antibodies made to order

The techniques for producing *human* monoclonal antibodies to predetermined specificities from hybridoma cells (cf. p. 122) still leave something to be desired and restlessly we look to recombinant DNA technology to satisfy the need (figure 14.1). There are several approaches to the production of antibodies which do not depend upon the human immune system, such as the Fab and single chain Fv (V_H - V_L) constructs (cf. p. 124) selected from phage libraries. Single V_H domain antibodies, being so small, may well be capable of reaching

cell receptors on viruses which are tucked away at the bottom of protein canyons where they might be inaccessible to the Fv of an intact antibody, but the sticky nature of these V_H domains has to be addressed. Some antibodies possess peptidase activity, particularly where the light chains have homology to serum proteases, and this could lead to enhanced degradation of the antigen. Expression of **antibody genes in plants** is going to be big business; the technology is now in place to produce recombinant IgA antibodies coupled to secretory pieces in plants, and these should provide an invaluable supplement to dried cows' milk baby food in cases where the mother's milk is of poor quality.

Never neglect innate immune mechanisms. Defensins, the broad-range antimicrobial peptides present in polymorphonuclear neutrophil (PMN) granules (cf. p. 9), are now being engineered in tobacco plants and it is planned to use them for fungal and bacterial infections which become refractory to conventional antibiotics. A good example of lateral thinking by the project leaders.

Adoptive transfer of cytotoxic T-cells

This is a labor-intensive operation and will be restricted to instances where the donor shares an MHC class I allele. To give one example, up to 30% of recipients of bone marrow allografts from mismatched family members or matched unrelated donors develop Epstein-Barr virus (EBV) lymphoma. Pilot studies aimed at potential prophylaxis showed that EBV-induced cytotoxic T-cell (Tc) lines transferred to the bone marrow recipients reconstituted the patients' immune responses to EBV for at least 18 months.

VACCINATION

Herd immunity

In the case of tetanus, active immunization is of benefit to the individual but not to the community since it will not eliminate the organism which is found in the feces of domestic animals and persists in the soil as highly resistant spores. Where a disease depends on human transmission, immunity in just a proportion of the population can help the whole community if it leads to a fall in the reproduction rate (i.e. the number of further cases produced by each infected individual) to less than one; under these circumstances the disease will die out: witness, for example, the disappearance of diphtheria from communities in which around 75% of the children have been immunized (figure 14.2). But

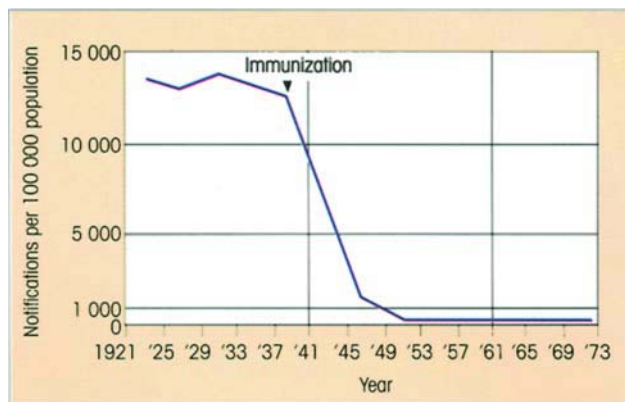


Figure 14.2. Notification of diphtheria in England and Wales per 100 000 population showing dramatic fall after immunization. (Reproduced from Dick G. (1978) *Immunisation*. Update Books; with kind permission of the author and publishers.)

this figure must be maintained; there is no room for complacency. In contrast, focal outbreaks of measles have occurred in communities which object to immunization on religious grounds, raising an important point for parents in general. Each individual must compare any perceived disadvantage associated with vaccination in relation to the increased risk of disease in their unprotected child.

Strategic considerations

The objective of vaccination is to provide effective immunity by establishing adequate levels of antibody and a primed population of memory cells which can rapidly expand on renewed contact with antigen and so provide protection against infection. Sometimes, as with polio infection, a high blood titer of antibody is required; in mycobacterial diseases, such as tuberculosis (TB), a macrophage-activating cell-mediated immunity (CMI) is most effective, whereas with influenza virus infection, cytotoxic T-cells probably play a significant role. The site of the immune response evoked by vaccination may also be most important. For example, in cholera, antibodies need to be in the gut lumen to inhibit adherence to and colonization of the intestinal wall.

Eradication of the infectious agent is not always the most practical goal. To take the example of malaria, the blood-borne form releases molecules which trigger tumor necrosis factor (TNF) and other cytokines from monocytes, and the secretion of these mediators is responsible for the unpleasant effects of the disease. Accordingly, an antibody response targeted to these

Table 14.2. Factors required for a successful vaccine.

FACTOR	REQUIREMENTS
Effectiveness	Must evoke protective levels of immunity: at the appropriate site of relevant nature (Ab, Tc, Th1,Th2) of adequate duration
Availability	Readily cultured in bulk or accessible source of subunit
Stability	Stable under extreme climatic conditions, preferably not requiring refrigeration
Cheapness	What is cheap in the West may be expensive in developing countries but WHO and others try to help
Safety	Eliminate any pathogenicity

released antigens with structurally conserved epitopes may be a realistic holding strategy, while the search for a global vaccine aimed at the more elusive antigen-swapping parasite itself is grinding forward. Under these circumstances life with the parasite might be acceptable.

In addition to an ability to engender effective immunity, a number of mundane but nonetheless crucial conditions must be satisfied for a vaccine to be considered successful (table 14.2). The antigens must be readily available, and the preparation should be stable, cheap and certainly, safe, bearing in mind that the recipients are most often healthy children. Clearly, the first contact with antigen during vaccination should not be injurious and the maneuver is to avoid the pathogenic effects of infection, while maintaining protective immunogens.

KILLED ORGANISMS AS VACCINES

The simplest way to destroy the ability of microbes to cause disease, yet maintain their antigenic constitution, is to prevent their replication by killing in an appropriate manner. Parasitic worms and, to a lesser extent, protozoa are extremely difficult to grow up in bulk to manufacture killed vaccines. This problem does not arise for many bacteria and viruses and, in these cases, the inactivated microorganisms have generally provided safe antigens for immunization. Examples are typhoid (in combination with the relatively ineffective paratyphoid A and B), cholera and killed poliomyelitis (Salk) vaccines. The success of the Salk vaccine was slightly marred by a small rise in the incidence of deaths from poliomyelitis in 1960–61 (figure 14.3), but this has now been attributed to poor antigenicity of one of the three different strains of virus used, and present-day vaccines are far more potent. Care has to be taken to ensure that important protec-

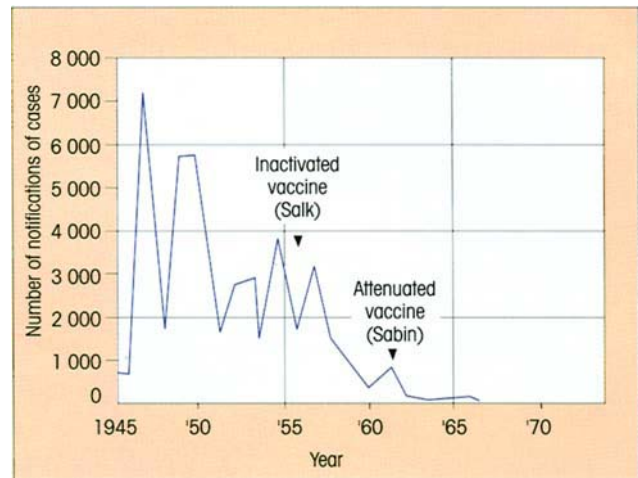


Figure 14.3. Notifications of paralytic poliomyelitis in England and Wales showing the beneficial effects of community immunization with killed and live vaccines. (Reproduced from Dick G. (1978) *Immunisation*. Update Books; with kind permission of the author and publishers.)

tive antigens are not destroyed in the inactivation process. During the production of an early killed measles vaccine, the fusion antigen, which permits cellular spread of virus, was inactivated; as a result, incomplete immunity was produced and this left the individual susceptible to the development of immunopathological complications on subsequent natural infection. The dangers of incomplete immunity are especially worrying in areas where measles is endemic and the immune response is relatively enfeebled due to protein malnutrition. Since the widespread correction of this dietary deficiency is unlikely in the near future, it is worth considering whether non-specific stimulation by immunopotentiating drugs or thymus hormones at the time of vaccination might provide a feasible solution.

LIVE ATTENUATED ORGANISMS HAVE MANY ADVANTAGES AS VACCINES

The objective of attenuation is to produce a modified organism which mimics the natural behavior of the original microbe without causing significant disease. In many instances the immunity conferred by killed vaccines, even when given with adjuvant (see below), is often inferior to that resulting from infection with live organisms. This must be partly because the replication of the living microbes confronts the host with a **larger and more sustained dose of antigen** and that, with budding viruses, infected cells are required for the establishment of good **cytotoxic T-cell memory**.

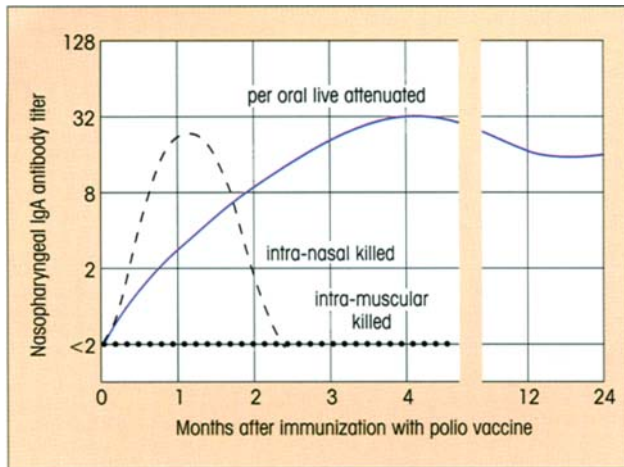


Figure 14.4. Local IgA response to polio vaccine. Local secretory antibody synthesis is confined to the specific anatomical sites which have been directly stimulated by contact with antigen. (Data from Ogra P.L. *et al.* (1975) In Notkins A.L. (ed.) *Viral Immunology and Immunopathology*, p. 67. Academic Press, New York.)

Another significant advantage of using live organisms is that the immune **response takes place largely at the site of the natural infection**. This is well illustrated by the nasopharyngeal IgA response to immunization with polio vaccine. In contrast with the ineffectiveness of parenteral injection of killed vaccine, intranasal administration evoked a good local antibody response; however, whereas this declined over a period of 2 months or so, per oral immunization with *live attenuated* virus established a persistently high IgA antibody level (figure 14.4).

There is in fact a strong upsurge of interest in strategies for mucosal immunization. Remember, the MALT system involves mucous membranes covering the aerodigestive and urogenital tracts as well as the conjunctiva, the ear and the ducts of all exocrine glands which are essentially protected by sIgA antibodies. Resident T-cells in these tissues produce large amounts of transforming growth factor- β (TGF β), and the interleukins IL-10 and IL-4, which promote the B-cell switch to IgA, and note also that human intestinal epithelial cells themselves are major sources of TGF β and IL-10.

Classical methods of attenuation

The objective of attenuation, that of producing an organism which causes only a very mild form of the natural disease, can be equally well attained if one can identify heterologous strains which are virulent for another species, but avirulent in humans. The

best example of this was Jenner's seminal demonstration that cowpox would protect against smallpox. Since then, a truly remarkable global effort by the World Health Organization (WHO), combining extensive vaccination and selective epidemiological control methods, **has completely eradicated the human disease**—a wonderful achievement. Emboldened by this success, the WHO embarked upon a program to eradicate polio and, despite setbacks caused by armed conflict limiting access to local populations, it is hoped to declare the world free of polio by 2005. One can even follow the progress of this campaign on <http://www.polioeradication.org>.

Attenuation itself was originally achieved by empirical modification of the conditions under which an organism grows. Pasteur first achieved the production of live but nonvirulent forms of chicken cholera bacillus and anthrax (cf. Milestone 14.1) by such artifices as culture at higher temperatures and under anerobic conditions, and was able to confer immunity by infection with the attenuated organisms. A virulent strain of *Mycobacterium tuberculosis* became attenuated by chance in 1908 when Calmette and Guérin at the Institut Pasteur, Lille, added bile to the culture medium in an attempt to achieve dispersed growth. After 13 years of culture in bile-containing medium, the strain remained attenuated and was used successfully to vaccinate children against tuberculosis. The same organism, BCG (bacille Calmette–Guérin), is widely used today for the immunization of tuberculin-negative individuals. Attenuation by cold adaptation of influenza and other respiratory viruses seems hopeful; the organism can grow at the lower temperatures (32–34°C) of the upper respiratory tract, but fails to produce clinical disease because of its inability to replicate in the lower respiratory tract (37°C).

Attenuation by recombinant DNA technology

It must be said that many of the classical methods of attenuation are somewhat empirical and the outcome is difficult to control or predict. With knowledge of the genetic makeup of these microorganisms, we can apply the molecular biologist's delicate scalpel to deliberately target the alterations in life-style which are needed for successful attenuation. Thus genetic recombination is being used to develop various attenuated strains of viruses, such as influenza, with lower virulence for humans and some with an increased multiplication rate in eggs (enabling newly endemic strains of influenza to be adapted for rapid vaccine production). Not surprisingly, strains of HIV-1, with vicious deletions of the regulatory genes, are being

investigated as protective vaccines. The potential is clearly quite enormous.

The **tropism** of attenuated organisms for the site at which **natural infection** occurs is likely to be exploited dramatically in the near future to establish gut immunity to typhoid and cholera using attenuated forms of *Salmonella* strains and *Vibrio cholerae* in which the virulence genes have been identified and modified by genetic engineering.

Microbial vectors for other genes

An ingenious trick is to use a virus as a 'piggy-back' for genes encoding a vaccine immunogen. Incorporation of such 'foreign' genes into attenuated recombinant viral vectors, such as fowlpox virus and modified vaccinia virus Ankara strain, which infect mammalian hosts but are unable to replicate effectively, provides a powerful vaccination strategy with many benefits. The genes may be derived from organisms which are difficult to grow or inherently dangerous, and the constructs themselves are replication deficient, nonintegrating, stable and relatively easy to prepare. The proteins encoded by these genes are appropriately expressed *in vivo* with respect to glycosylation and secretion, and are processed for major histocompatibility complex (MHC) presentation by the infected cells, thus effectively endowing the host with both humoral immunity and CMI.

A wide variety of genes have been expressed by vaccinia virus vectors, and it has been demonstrated that the products of genes coding for viral envelope proteins, such as influenza virus hemagglutinin, vesicular stomatitis virus glycoprotein, human immunodeficiency virus (HIV)-1 gp120 and herpes simplex virus glycoprotein D, could be correctly processed and inserted into the plasma membrane of infected cells. Hepatitis B surface antigen (HBsAg) was secreted from recombinant vaccinia virus-infected cells as the characteristic 22 nm particles (figure 14.5). It is an impressive approach and chimpanzees have been protected against the clinical effects of hepatitis B virus, while mice inoculated with recombinant influenza hemagglutinin generated cytotoxic T-cells and were protected against influenza infection.

Attention has turned to BCG as a vehicle for antigens required to evoke CD4-mediated T-cell immunity. The organism is avirulent, has a low frequency of serious complications, can be administered any time after birth, has strong adjuvant properties, gives long-lasting CMI after a single injection and is a bargain at around US \$0.05 a shot. The development of shuttle vectors which can replicate in *E. coli* as plasmids and in

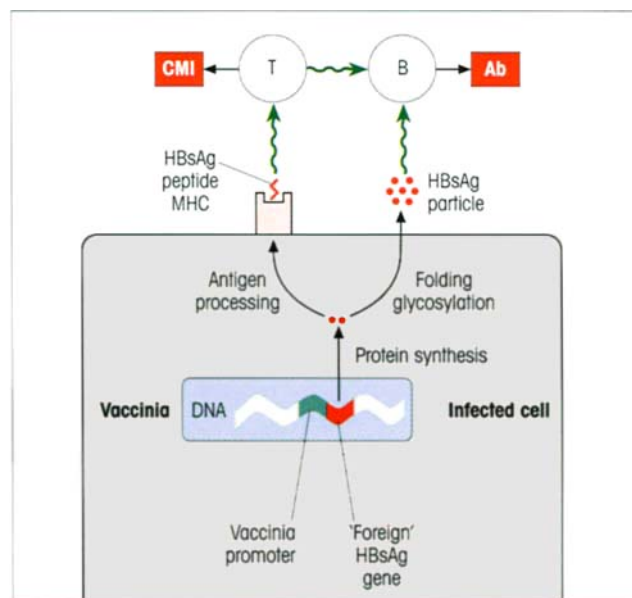


Figure 14.5. Hepatitis B surface antigen (HBsAg) vaccine using an attenuated vaccinia virus carrier. The HBsAg protein is synthesized by the machinery of the host cell: some is secreted to form the HBsAg 22 nm particle which stimulates antibody (Ab) production, and some follows the antigen processing pathway to stimulate cell-mediated immunity (CMI) and T-helper activity.

mycobacteria as phages has allowed foreign DNA to be introduced into *M. smegmatis* and BCG vaccine strains. We can expect many advances on this front: thus incorporation of a gene for kanamycin resistance into the plasmid provides a selectable marker for transformed bacteria, while inclusion of a signal sequence permits secretion of the recombinant protein. Recombinant BCG, engineered to express the outer surface protein A (OspA) of *Borrelia* in its cell membrane, induced excellent antibody titers. For those interested, *Borrelia burgdorferi* is the cause of Lyme disease associated with an arthritic condition.

The ability of *Salmonella* to elicit **mucosal responses by oral immunization** has been exploited by the design of vectors which allow the expression of any protein antigen linked to *E. coli* enterotoxin, a powerful mucosal immunostimulant. There is an attractive possibility that the oral route of vaccination may be applicable not only for the establishment of gut mucosal immunity but also for providing systemic protection. For example, *Salmonella typhimurium* not only invades the mucosal lining of the gut, but also infects cells of the mononuclear phagocyte system throughout the body, thereby stimulating the production of humoral and secretory antibodies as well as CD4 and CD8 cell-mediated immunity. Since attenuated *Salmonella* can be made to express proteins from *Shigella*, cholera, malaria sporozoites and so on, it is entirely feasible to

consider these as potential oral vaccines. *Salmonella* may also carry 'foreign genes' within separate DNA plasmids and, after phagocytosis by antigen-presenting cells, the plasmids can be released from the phagosome into the cytosol if the plasmid bears a recombinant lysteriolysin gene or the bacterium is an auxotrophic mutant whose cell walls disintegrate within the phagosome; the plasmid then moves to the nucleus where it is transcribed to produce the desired antigen. Quite strikingly, these attenuated organisms are very effective when inhaled; for example, intranasal immunization with recombinant BCG expressing the OspA lipoprotein (*vide supra*) elicited substantive mucosal and systemic immune responses comparable to those obtained by the parenteral route. Vaccinologists are still confidently predicting 'the age of the nose' and we should soon hear more of human trials with this wide variety of attenuated vectors.

Constraints on the use of attenuated vaccines

Attenuated vaccines for poliomyelitis (Sabin), measles and rubella have gained general acceptance. However, with live viral vaccines there is a possibility that the nucleic acid might be incorporated into the host's genome or that there may be reversion to a virulent form, as seen recently with the reactivation of the live attenuated form of varicella zoster virus (chickenpox) in individuals with a relatively low titer response. Reversion is less likely if the attenuated strains contain several mutations. Another disadvantage of attenuated strains is the difficulty and expense of maintaining appropriate cold-storage facilities, especially in out-of-the-way places. In diseases such as viral hepatitis, AIDS and cancer, the dangers associated with live vaccines are daunting, as witnessed by the experience with attenuated simian immunodeficiency virus in monkeys (see p. 319). As discussed above, with certain vaccines there is a very small, but still real, risk of developing complications and it cannot be emphasized too often that this **risk must be balanced against the expected chance of contracting the disease with its own complications**. Where this is minimal, some may prefer to avoid general vaccination and to rely upon a crash course backed up if necessary by passive immunization in the localities around isolated outbreaks of infectious disease.

It is important to recognize those children with immunodeficiency before injection of live organisms; a child with impaired T-cell reactivity can become overwhelmed by BCG and die. Perhaps this is only a sick story, but it is said that in one particular country at a certain time there were no adults with T-cell deficiency.

The reason? All children had been immunized with live BCG as part of a community health program(!). The extent to which children with partial deficiencies are at risk has yet to be assessed. It is also inadvisable to give live vaccines to patients being treated with steroids, immunosuppressive drugs or radiotherapy or who have malignant conditions such as lymphoma and leukemia; pregnant mothers must also be included here because of the vulnerability of the fetus.

Use in a veterinary context

For veterinary use, of course, there is a little less concern about minor side-effects and excellent results have been obtained using existing vaccinia strains with rinderpest in cattle and rabies in foxes, for example. In the latter case, a recombinant vaccinia virus vaccine expressing the rabies surface glycoprotein was distributed with bait from the air and immunized approximately 80% of the foxes in that area. No cases of rabies have since been seen, but epidemiological considerations indicate that, with the higher fox density that this leads to, the higher the percentage which have to be made immune; thus, either one has to increase the efficacy of the vaccine, or culling of the animals must continue—an interesting consequence of interference with ecosystems. Less complicated is the use of such immunization to control local outbreaks of rabies in rare mammalian species, such as the African wild dog in certain game reserves which are threatened with extinction by the virus.

SUBUNIT VACCINES CONTAINING INDIVIDUAL PROTECTIVE ANTIGENS

A whole parasite or bacterium usually contains many antigens which are not concerned in the protective response of the host but may give rise to problems by suppressing the response to protective antigens or by provoking hypersensitivity, as we saw in the last chapter. Vaccination with the isolated protective antigens may avoid these complications, and identification of these antigens then opens up the possibility of producing them synthetically in circumstances where bulk growth of the organism is impractical or isolation of the individual components too expensive.

Identification of protective antigens is greatly facilitated if one has an experimental model. If protection is antibody-mediated, one can try out different monoclonal antibodies and use the successful ones to pull out the antigen. Where antigenic variation is a major factor, desperate attempts are being made to identify some element of constancy which could provide a

basis for vaccination, again using monoclonal antibodies with their ability to recognize a single specificity in a highly complex mixture. If protection is based primarily on T-cell activity, the approach would then be through the identification of individual T-cell clones capable of passively transferring protection. Switching back to humans, one seeks encouragement that the experimental models have kept the focus on the right target by confirming that the immune response to the antigen identified in the models correlates with protection in naturally infected individuals.

The use of purified components

Bacterial exotoxins such as those produced by diphtheria and tetanus bacilli have long been used as immunogens. First, they must of course be detoxified and this is achieved by formaldehyde treatment, which fortunately does not destroy the major immunogenic determinants (figure 14.6). Immunization with the **toxoid** will therefore provoke the formation of protective antibodies, which neutralize the toxin by stereochemically blocking the active site, and encourage removal by phagocytic cells. The toxoid is generally given after adsorption to aluminum hydroxide which acts as an adjuvant and produces higher antibody titers.

The emphasis now is to move towards gene cloning of individual proteins once they have been identified immunologically and biochemically. In general, a protein subunit used in a vaccine should contain a sufficient number of T-cell epitopes to avoid human leukocyte antigen (HLA)-related unresponsiveness within the immunized population. In order to maintain a pool of memory B-cells over a reasonable period of time, persistence of antigen on the follicular dendritic cells in a form resistant to proteolytic degradation with retention of the native three-dimensional configuration is needed. Glycosylation of the protein contributes to this stability, but by the same token might not give a good T-cell response, so that the vaccine may need to be supplemented with a separate

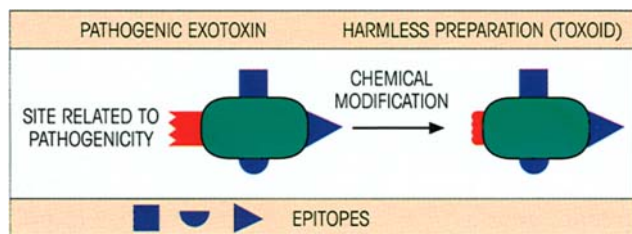


Figure 14.6. Modification of toxin to harmless toxoid without losing many of the antigenic determinants. Thus antibodies to the toxoid will react well with the original toxin.

denatured source of T-cell epitopes. Purified polysaccharide vaccines are in a different category in that they normally require coupling to some immunogenic carrier protein, such as tetanus toxoid or mycobacterial heat-shock protein (figure 14.7), since they fail to stimulate T-helpers or induce adequate memory. This maneuver can give respectable antibody titers, but these will only be boosted by a natural infection if the carrier is derived from or related to the infecting agent itself.

Antigens can be synthesized through gene cloning

Recombinant DNA technology enables us to make genes encoding part or the whole of a protein peptide chain almost at will, and express them in an appropriate vector. We have already ruminated upon vaccinia virus and other recombinant vectors. Another strategy is to fuse the gene with the Ty element of yeast which self-assembles into a highly immunogenic virus-like particle. In a similar fashion, peptides can be fused with the core antigen of hepatitis B virus, which spontaneously polymerizes into 27 nm particles capable of eliciting strong T-cell help. However, we often wish to develop vaccines which utilize the gene product on its own, incorporated in an adjuvant. Baculovirus vectors in moth cell lines produce large amounts of glycosylated recombinant protein, while the product secreted by yeast cells expressing the *HBsAg* gene is available as a commercial vaccine. Stably transformed transgenic bananas are now being developed to express

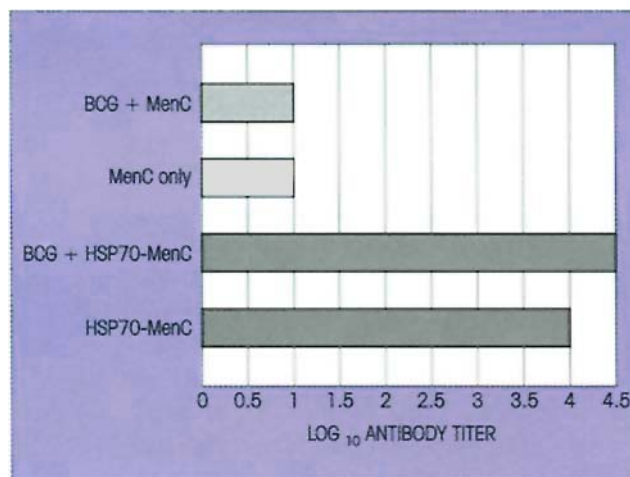


Figure 14.7. The carrier effect of mycobacterial heat-shock protein (hsp70) without adjuvant. Antibody responses to group C meningococcal polysaccharide (MenC) conjugated to hsp70 injected into mice with and without priming to the attenuated *Mycobacterium* BCG. (From Lambert P.-H., Louis J.A. & del Giudice G. (1992) In Gergely *et al.* (eds) *Progress in Immunology* VIII, pp. 683–689. Springer-Verlag, Budapest.)

protein vaccines. Bananas are cheap and easy to grow in the developing world and children like to eat them raw so avoiding inactivation by cooking. It is a sobering thought that a few hectares of the fruit could satisfy the annual global requirement for a single dose of oral immunogen.

The potential of gene cloning is clearly vast and, in principle, economical, but there are sometimes difficulties in identifying a good expression vector, in obtaining correct folding of the peptide chain to produce an active protein, and in separating the required product from the culture *mélange* in an undenatured state. One restriction is that carbohydrate antigens cannot be synthesized directly by recombinant DNA technology, although many of the cohort of genes which encode the cascade of enzymes needed to produce complex carbohydrates have themselves now been cloned.

The naked gene itself acts as a vaccine

Teams working with J. Wolff and P. Felgner experimented with a new strategy for gene therapy which involved binding the negatively charged DNA to cationic lipids which would themselves attach to the negatively charged surface of living cells and then presumably gain entry. The surprise was that controls injected with DNA without the lipids actually showed an *even higher uptake of DNA* and expression of the protein it encoded, so giving rise to the whole new technology of **naked DNA therapy**. As Wolff put it: 'We tried it again and it worked. By the fourth or fifth time we knew we were onto something big. Even now I get a chill down my spine when I see it working.' Well, there is the real excitement of a blockbuster finding, even if, as usually happens to be the case, it arises from serendipity. It was quickly appreciated that the injected DNA functions as a source of immunogen *in situ* and can induce strong immune responses. So, now, vaccinologists everywhere are scurrying around trying to adapt the new technology.

The transcription unit composed of the cDNA gene with a poly A terminator is stitched in place in a DNA plasmid with a cytomegalovirus promoter and a CpG bacterial sequence as a very potent adjuvant. It is usually injected into muscle where it can give prolonged expression of protein. Undoubtedly, the pivotal cell is the dendritic antigen-presenting cell which may be transfected directly, could endocytose soluble antigen secreted into the interstitial spaces of the muscle, and could take up cells that have been killed or injured by the vaccine. The CpG immunostimulatory sequences provoke the synthesis of IFN α and β , IL-12 and IL-18 which promote the formation of Th1 cells; this in turn

generates good cell-mediated immunity, helps the B-cell synthesis of certain antibody classes (e.g. IgG2a in the mouse) and induces good cytotoxic T-cell responses, presumably reflecting the cytosolic expression of the protein and its processing with MHC class I. Let's look at an example. It will be recalled that frequent point mutations (drift; p. 267) in antigenically important regions of influenza surface hemagglutinin give rise to substantial antigenic variation, whereas the major internal proteins which elicit T-cell-mediated immunity responses have been relatively conserved over the last 60 years. On this line of reasoning, nucleoprotein DNA should give broad protection against other influenza strains and indeed it does (figure 14.8). A combination of DNAs encoding the hemagglutinin (included only for statutory reasons) and nucleoprotein genes gave nonhuman primates and ferrets good protection against infection, and protected ferrets against challenge with an antigenically distinct epidemic human virus strain more effectively than the contemporary clinically licensed vaccine. Excellent antibody responses to the viral hemagglutinin were also readily obtained in monkeys. Vaccination can also be achieved by coating the plasmids onto minute gold particles and shooting them into skin cells by the high-pressure 'biolistic' helium gun (cf. p. 141). This technique uses 100-fold less plasmid DNA than the muscle injection and the response is skewed more towards Th2-dependent antibody production, possibly because the antigen dose is too low to foster Th1 cells.

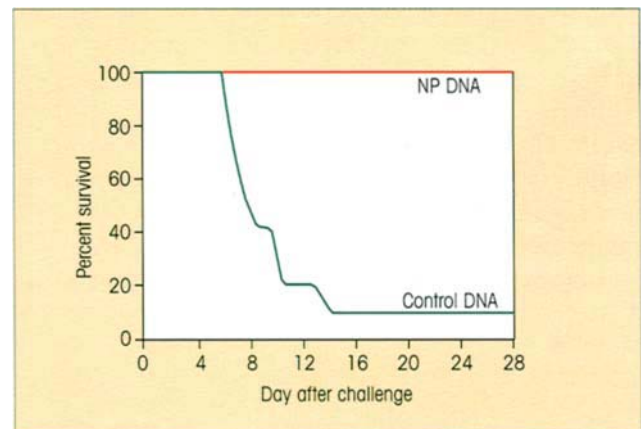
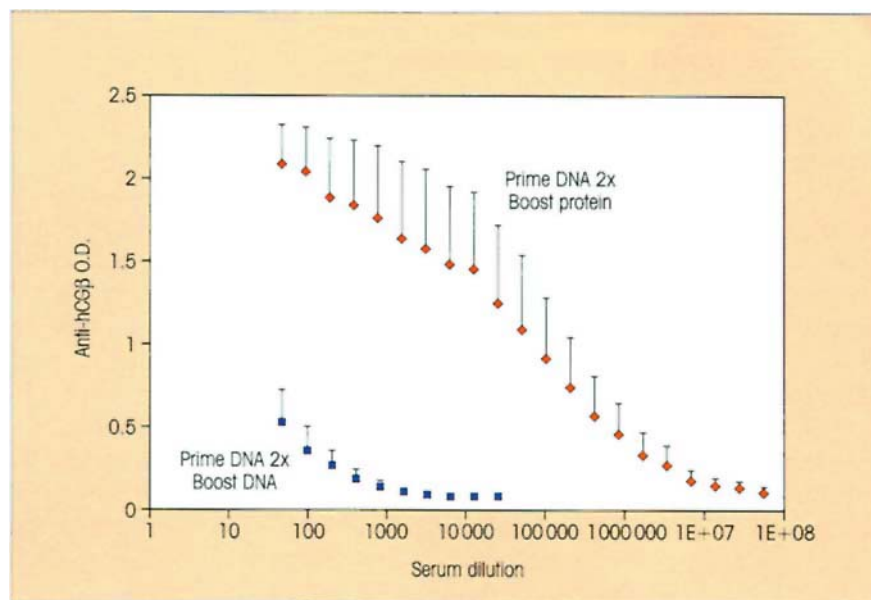


Figure 14.8. Protection from cross-strain influenza challenge after vaccination with nucleoprotein DNA. Mice were immunized three times at 3-week intervals with 200 μ g of nucleoprotein (NP) or vector (control) DNA and lethally challenged with a heterologous influenza strain 3 weeks after the last immunization. Survival of mice given NP DNA was significantly higher than in mice receiving vector ($p=0.0005$). (Data kindly provided by Dr Margaret A. Liu and colleagues from *DNA and Cell Biology* (1993) 12(9), and reproduced by permission of Mary Ann Liebert Inc.)

Figure 14.9. Induction of memory cells by DNA vaccine and boost of antibody production with the protein immunogen, human chorionic gonadotropin- β (β -hCG). Groups of five (C57BL/6 $\times$ BALB/c)F1 mice each received 50 μ g of the β -hCG DNA plasmid at weeks 0 and 2; one group received a further injection of the plasmid, while the other was boosted with 5 μ g of the β -hCG protein antigen in RIBI (cf. p. 300) adjuvant. Dilutions of serum were tested for antibodies to β -hCG by indirect ELISA. Mean titers $\pm$ SE are shown. (Data from Laylor R., Lund T. *et al.* (1999) *Clinical and Experimental Immunology* 117, 106.)



The persistent but low level of expression of the protein antigen by naked DNA vaccines establishes a pool of relatively high affinity memory B-cells which can readily be revealed by boosting with protein antigen (figure 14.9). This has given rise to a rather impressive ‘**prime and boost**’ strategy where these memory cells are expanded by boosting with a nonreplicating viral vector, such as fowlpox virus or Ankara strain-modified vaccinia virus, bearing a gene encoding the antigen. Mice immunized in this fashion with influenza virus hemagglutinin produced satisfyingly high levels of IgG2a antibody and were protected against challenge with live virus. Remarkably, up to 30% of circulating CD8 T-cells were specific for the immunizing epitope as shown by MHC class I tetramer binding (cf. p. 138). A similar strategy with *Plasmodium berghei* produced high levels of peptide-specific CD8 cells secreting IFN γ which protected against challenge by sporozoites.

A new generation of genetic vaccines overcomes the poor efficacy of some DNA- and RNA-based vaccines by incorporating the replicating machinery used by members of the alphavirus genus, which includes the Sindbis and Semliki Forest viruses, into the DNA plasmid. The **alphaviral genome** consists of a single positive-stranded RNA encoding structural genes, one of which can be substituted by the antigen gene, and an RNA replicase. Cells transfected with the plasmid replicon become loaded with alphavirus and antigen and meet a sticky apoptotic end whereupon they are taken up by antigen-presenting cells to initiate a powerful immune response.

There is a lot going for DNA vaccines. Considering influenza virus, for example, the DNA needed to prepare a vaccine can be obtained directly from current clinical material without having to select specific mutant strains. Altogether, the speed and simplicity mean that the 2 years previously needed to make a recombinant vaccine can be reduced to months. DNA vaccines do not need the cumbersome and costly protein synthesis and purification procedures that subunit formulations require; almost identical production facilities can be used for totally different vaccine candidates; and they can be prepared in a highly stable powder form which does not depend on the cold complex chain logistically needed for heat-sensitive vaccines, such as the oral polio vaccine in tropical countries. But, above all, they are incredibly cheap. The \$3 required for an injection of recombinant hepatitis B represents the whole health budget for a single individual in some countries, whereas the single shot of DNA vaccine would be a tiny fraction of that. Current trials are addressing a number of safety considerations, such as the possibility of permanent incorporation of a plasmid into the host genome. The widespread use of DNA vaccines in humans will depend upon the results of such trials, although observations to date have generally been very encouraging.

EPITOPE-SPECIFIC VACCINES MAY BE NEEDED

Most immunogens, especially proteins, present a variety of B- and T-cell epitopes to the immune system.

Most, if not all, will elicit protective responses, but some may have undesirable characteristics. For example, if there is cross-reaction with a self-epitope, as between *Trypanosoma cruzi* and heart muscle, potentially pathogenic autoimmune reactions may result. Sometimes an immunodominant region, such as the V3 loop in HIV gp120, hogs the antibody response at the expense of the more weakly immunogenic conserved regions, but continually escapes from capture by a high mutation rate. Similar escape mutants crop up in the Tc response to highly mutating dominant T-cell epitopes in malaria and various viral protein antigens. Unwanted epitope effects are seen in the phenomenon of 'original antigenic sin', in which a second infection with influenza virus involving an antigenically related but not identical strain generates antibodies with a higher titer for the strain which produced the first infection. It is conceivable also that certain epitopes may bias T-helpers towards an inappropriate subset or may engender a predominantly suppressive, antagonistic or anergic response which could downregulate T-cell reactivity to linked epitopes on the same immunogen (figure 14.10).

Frequently, a natural infection gives rise to a poorly functional protective antibody response and discourages the search for vaccine candidates which could

generate effective humoral immunity. Nonetheless, in many of these infections, a deliberately selected monoclonal antibody specific for more weakly immunogenic conserved regions of the microbe can be protective, as has been demonstrated in models of HIV, Ebola virus and *Candida* infections. In other words, the epitope giving rise to the protective monoclonal antibody is subdued as an immunogen when present as a component of the infectious microbe, but potentially might be exploited to stimulate high levels of protective antibodies if it could be extracted from the molecular 'woodwork' so to speak.

Bearing in mind the many different circumstances we have discussed, the requirement may arise for a vaccine in which the good protective epitopes can be brought into the front line and dissociated from the molecular environment of the 'bad guys' which compromise the protective responses. In other words, we need to construct **epitope-specific vaccines**, preferably based on conserved regions, which provide broad defense. Several approaches are possible.

Epitopes can be mimicked by synthetic peptides

B-cell epitopes

Small peptide sequences corresponding to important epitopes on a microbial antigen can be synthesized readily and economically; long ones are rather expensive to manufacture. One might predict that, although the synthetic peptide has the correct linear *sequence* of amino acids, its random structure would make it a poor model for the *conformation* of the parent antigen and hence a poor vaccine for evoking humoral immunity. Curiously, this does not always seem to be a serious drawback. The 20-amino acid peptide derived from the foot and mouth virus-specific protein (VP1) evokes a good neutralizing response. The explanation has been forthcoming from X-ray structural analysis which shows the peptide sequence to be in a 'loop' region with blurred electron density indicative of dramatic disorder. In this case, the epitope is linear and evidently the flexibility of the loop structure may approach that of the free peptide which can thus mimic the epitope on the native VP1 molecule and stimulate a protective antibody response when used as a vaccine (figure 14.11a and b). Where the epitope is linear but is restricted in conformation by adjacent structures in the intact protein, immunization with free peptide tends to produce antibodies of disappointing affinity for the protein itself for the reasons outlined in figure 14.11c.

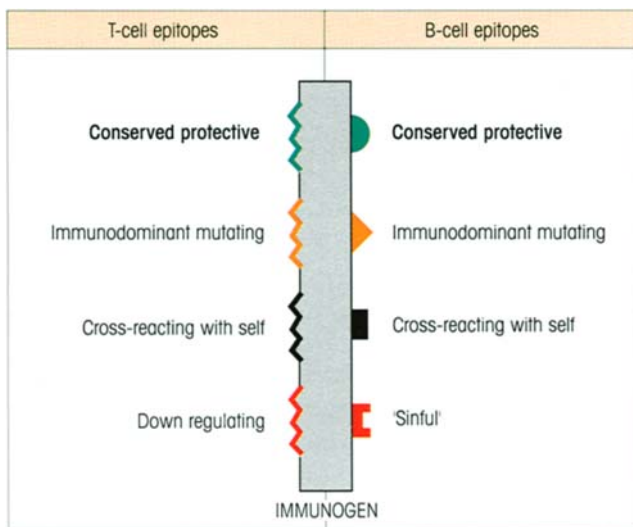


Figure 14.10. Desirable (green) and unwanted (red) epitopes on a protein immunogen. Conserved epitopes give broader protection against mutant variant strains although they may sometimes be weakly immunogenic when present in the whole microbe. Epitopes may be unwanted because: (i) they are immunodominant and attract the main immune response but continually escape by mutation; (ii) they cross-react with self-epitopes and can trigger autoimmunity; (iii) they are responsible for 'original antigenic sin' (see text); or (iv) they downregulate or antagonize T-cell-mediated immunity.

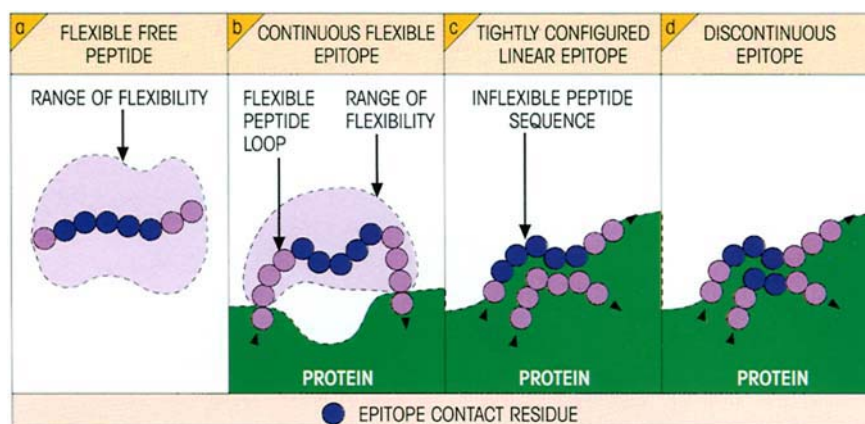


Figure 14.11. Structural basis for peptide mimicry of protein epitopes. (a) The free peptide is very flexible and can adopt a large number of structures in solution. (b) If the peptide sequence is present as a linear epitope on a part of a protein which is a flexible loop or chain, this will also exist in a variety of structures resembling the free peptide to a fair extent, and will behave comparably as an antigen and as an immunogen (vaccine) so that the peptide will raise antibodies which react well with the native protein. (c) If the linear epitope on the protein is structurally constrained (i.e. inflexible), it represents only one of the many structures adopted by the free peptide; thus if this peptide is used for immunization, only a minority of the B-cells stimulated will be complementary in shape to the native protein, so the peptide would be a poor vaccine for humoral immunity to microbes containing the protein antigen. (Note, however, that the antibodies produced would be good for Western immunoblots where the pro-

tein has been denatured after sodium dodecyl sulfate (SDS) treatment and the peptide structure is relatively free.) Preformed antibodies to the protein would react with the peptide, albeit with lower affinity, because energy must be used to constrain the peptide to the one structure which fits the antibody—just like the force used to restrain a madman in a strait-jacket. Where the sequence has a comparable degree of constraint in both peptide and protein, as in the disulfide-bonded loops in diphtheria toxin and hepatitis B surface antigen, anti-peptide sera react reasonably well with the native protein. (d) Most commonly, the epitopes are discontinuous and, even if with difficulty we can predict the contact residues, the techniques for designing a peptide with appropriate structure are not robust, although some progress is being made using antibody to select from a random bacteriophage library in which the peptides are constrained on a structural scaffold, such as that supporting the Ig CDR3 loop.

T-cell epitopes

Although short peptides may not have the conformation to stimulate adequate B-cell responses, they can prime antigen-specific T-cells which recognize the primary sequence rather than the tertiary configuration of the protein. If the primed T-cells mediate CMI and possibly act to help B-cells make antibody, they could enable the host to have a head start in mounting an effective response on subsequent exposure to natural infection, and this would prove to be a useful prophylactic strategy.

We have already alluded to T-cell epitopes, often dominant and subject to high mutation rates, which can downregulate or subvert protective CMI responses. Under these circumstances, conserved peptide sequences which form **subdominant or cryptic epitopes** (cf. p. 201) **can function as effective vaccines**. The ineffectiveness of these sequences in providing adequate MHC–peptide levels to *prime* resting T-cells when the whole protein is processed by antigen-presenting cells can be side-stepped by immunizing with adequate doses of the preformed synthetic peptide. Because **primed**, as compared with **resting**, T-cells can be stimulated by much lower concentrations

of MHC–peptide and do not necessarily require major costimulatory signals, they will react with infected cells which have processed and presented the cryptic epitopes; furthermore, because the primed T-cells will be directed against conserved sequences, they will therefore provide broad CMI protection against mutated strains.

A major worry about peptides as T-cell vaccines is the variation in ability to associate with the different polymorphic forms of MHC molecules present in an outbred population, which contributes to the immune response (*Ir*) gene effect described earlier (see p. 213). This is not quite as serious as it might be owing to groupings of HLA types with consensus structures of the main B and F pockets in the peptide-binding groove, enabling them to recognize very similar peptide motifs. So far, three of these so-called **HLA super-types**, linked to HLA-A2, -A3 and -B7 respectively, have been identified, covering in total around 90% of the population. Residues 378–398 of the malaria circumsporozoite protein provide an exception in being virtually a universal T-cell epitope recognizable by all individuals so far tested. Other examples are residues 307–319 in influenza hemagglutinin and both 830–843 and 947–967 in tetanus toxin. These sequences, and

possibly the highly conserved heat-shock proteins, can provide promiscuous (i.e. HLA-independent) T-cell helper epitopes to conjugate with peptide vaccines.

Making the peptides immunogenic

The immunogenicity of peptides for B-cells is invariably bound up with a dependence on T-cell help, and failure to provide linked T-cell epitopes is thought to be responsible for poor antibody responses to the foot and mouth disease VP1 loop peptide in cattle and pigs, and to polymers of the tetrapeptide asparagylalanyl-asparagylproline (NANP) of malarial circumsporozoite antigens in humans (cf. p. 297). When the general T-cell carrier, peptide 378–398 (see above), was coupled to (NANP)₃ tetramer repeat, good antibody responses were obtained in all strains of mice tested. Furthermore, after priming with this synthetic peptide, whole sporozoites would boost antibody titers. This brings up two points: first, as we have already noted, in order for the natural infection to boost, the T- and B-cell epitopes must both be present and, second, they must be linked so that the T-cell epitope is taken up for processing by the lymphocyte which recognizes the B-cell epitope (figure 9.12). This does not always imply that the link in the infectious agent must be covalent, since mice primed with the core antigen of hepatitis B virus gave excellent responses to the surface antigen when challenged with whole virions, i.e. an interstructural relationship may function in this regard as well as an intramolecular one. In contrast, animals immunized with an HBs B-cell peptide coupled with a streptococcal peptide T-cell carrier require boosting with the original vaccine but do not receive a boost from natural infection. Mycobacterial heat-shock proteins are excellent carriers for peptides even in the absence of adjuvants and irrespective of BCG priming (figure 14.7), possibly due to the preprogramming of responses to cross-reacting self-heat-shock proteins in the ‘immunological homunculus’ (cf. p. 208). In some cases, responses induced by protein carriers might suppress the ability of a second injection of the vaccine to boost peptide antibody levels through antigenic competition and, in this respect, peptides providing the carrier T-cell epitope have a distinct advantage. It is worth noting that the presentation of a peptide, such as the foot and mouth disease virus VP1 loop, in the form of an octamer coupled to a poly-L-lysine backbone produces responses of far greater magnitude than the monomer, a strategy which has proved successful when multiple clusters of peptides are linked to a small central oligolysine core in what has been labeled ‘the multiple antigen peptide (MAP)

system’. Are not the multiple peptide units acting as carriers for each other?

Notwithstanding these considerations, **the majority of protein determinants are discontinuous**, i.e. involve amino acid residues far apart in the primary sequence but brought close to each other by peptide folding (figure 14.11d; cf. p. 293). In such cases, peptides which represent linear sequences of the primary structure will, at best, only mimic part of a determinant and will generate low affinity responses. Defining a discontinuous determinant by X-ray crystallography and site-related mutagenesis takes a long time and, by computer analysis, perhaps even longer. Even when armed with this information, synthesis of a configured peptide which will topographically mimic the contact residues that constitute such an epitope still remains a serious challenge. Progress is being made in using monoclonal antibodies to select peptide epitopes binding with higher affinity from bacteriophage libraries of random hexa- or heptapeptides, which are constrained on a structural scaffold such as that holding the CDR3 loop in the immunoglobulin variable region. Well, we have already encountered this type of epitope. Since it would be selected by an antibody we would class it as an **anti-idiotypic** and, likewise, if we used it as an immunogen it would be acting as an **idiotypic**.

Anti-idiotypes can be exploited as epitope-specific vaccines

In principle, **internal image anti-idiotypes** provide surrogates for discontinuous B-cell epitopes through possession of an innate structure capable of binding with the antigen combining site of the idiotype antibody. Knowledge of the contact residues within the epitope is not necessary. Since antigen–antibody binding is wholly dependent upon noncovalent intermolecular forces, which only become significant when the interacting molecular surfaces are very close together (cf. p. 85), all that is required is that, from the spectrum of millions of different antibody shapes available, one selects those that have a closely complementary three-dimensional topographical configuration which facilitates the intimate contact needed for these forces to permit effective binding. If the reader trundles back to figure 11.10, it will be seen that there are two main categories of anti-idiotypic: Ab_{2α}, which recognize cross-reacting, presumably regulatory, idiotypes, and Ab_{2β}, which are thought to behave as internal images of the antigenic determinant. In those instances in which the major Id is mostly associated with a given specificity (e.g. murine T15 Id is largely present

on antibodies to phosphorylcholine), then $Ab_{2\alpha}$ anti-Ids could provide useful potential vaccines, particularly for carbohydrates which are notoriously poor immunogens in the very young. They might also be applicable to the expansion of specific antitumor clones with their own private idiotypes in cancer patients. In general, however, the internal image $Ab_{2\beta}$ anti-idiotypes are capable of stimulating a wider range of lymphocytes and must be the strategy of choice. Figure 14.12 maps out how to go about generating $Ab_{2\alpha}$ anti-idiotypic vaccines which mimic epitopes defined by high affinity monoclonal antibodies (i.e. idiotypes). We would call this **reverse immunization**, in the sense that we are reversing the normal progression from antigen to antibody by using antibody to generate antigen. The approach is eminently feasible, and a number of examples of mimicry of epitopes on molecules such as hepatitis B surface antigen and yeast killer toxins have been reported.

B-cell lymphomas each produce a single characteristic immunoglobulin whose hypervariable regions represent unique targets for immunotherapy. How encouraging to read that, after many years of toil, protective immunity against a surface Ig-negative myeloma has been achieved by injection of a DNA vaccine with the myeloma Ig scFv linked to fragment C of tetanus toxoid, which acts as a danger signal to stimulate the anti-Id response. Curiously, the effectors appeared to be CD4 rather than CD8 T-cells.

Mutants which have lost unwanted epitopes can correctly fold desired discontinuous B-cell epitopes

The most natural way to achieve a correctly configured discontinuous B-cell epitope is to allow the protein to fold spontaneously. If the gene encoding the protein antigen is mutated so that the unwanted epitope is eliminated by replacement of its amino acid side-chains, without affecting the folding of the protein chain which generates the epitopes we wish to preserve, our object is achieved. Preservation of the desired epitopes and destruction of the 'bad' epitopes by 'genetic sandpapering' can obviously be monitored by following the reactivity of the mutants with the appropriate monoclonal antibody (figure 14.13). It will be apparent that 'bad' T-cell epitopes can also be eliminated by targeted mutations.

Sometimes, it should be possible to mask an undesirable epitope by introducing a glycosylation site into the sequence. In other cases, one can eliminate large unwanted segments of the antigen if the remaining peptide sequence can still fold correctly. Thus, immunization with the highly conserved extracellular domain of the influenza virus M2 protein fused to the N-terminus of the hepatitis B core protein gave 90–100% protection against lethal viral challenge and, bearing in mind the conserved nature of the antigen, this ought to cover a variety of influenza strains.

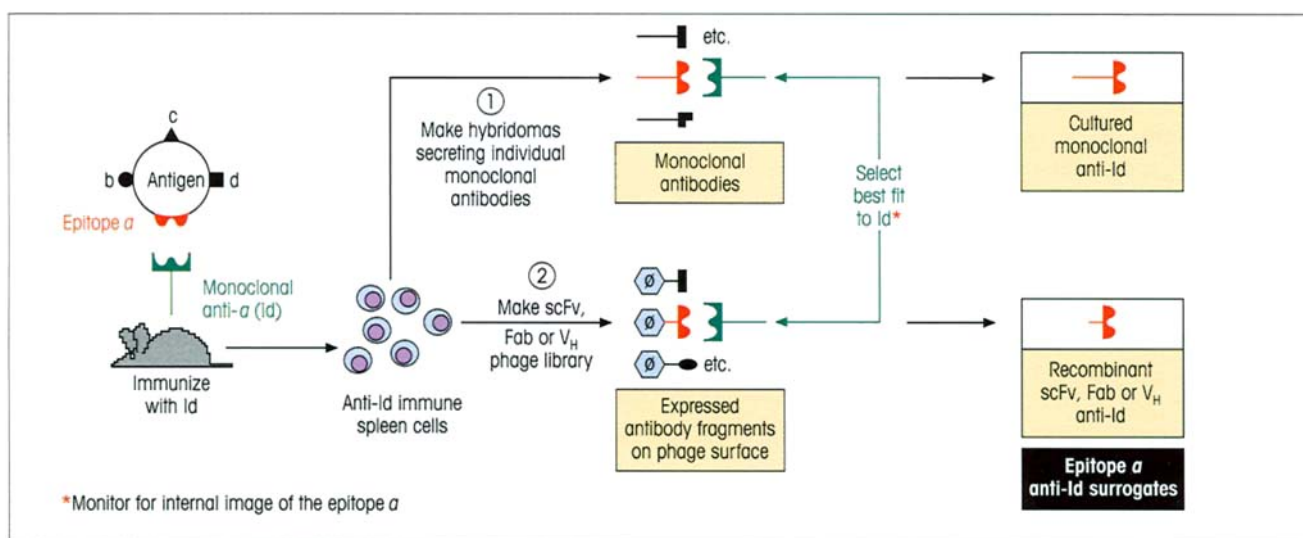


Figure 14.12. Derivation of anti-idiotypic mimics of a B-cell epitope. A high affinity monoclonal antibody (idiotype; Id) specific for the *a* epitope on the antigen is injected to generate an anti-Id response. Spleen cells from the immunized mice are used (1) to make a range of hybridomas, each secreting an individual monoclonal antibody, a small proportion of which will be internal image anti-Id, or (2) a phage library expressing surface antibody fragments. These are

screened with the original monoclonal Id to select the best fitting anti-Id monoclonals or antibody fragments. These, in turn, are monitored for behavior as internal image of *a* through ability to block binding of the original Id to *a*, to bind to other anti-*a* monoclonals and to be recognized by a polyclonal antiserum raised against the antigen in other species. A successful candidate can then be produced in bulk to provide a surrogate vaccine for epitope *a*.

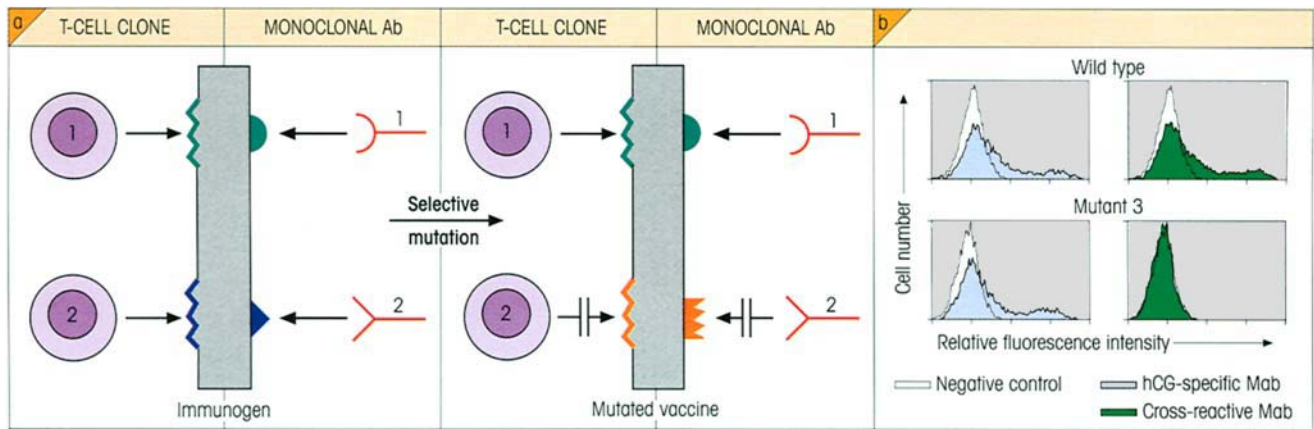


Figure 14.13. (a) Selective mutation of unwanted (dark blue) with retention of desirable (green) epitopes in a protein vaccine. Success of the mutation strategy can be monitored by reactivity with monoclonal antibodies (B-epitopes) and T-cell clones (T-epitopes). Our immunoprotein engineering group at University College London has used this strategy to mutate human chorionic gonadotropin- β (β -hCG) so that it still retains β -hCG-specific epitopes but has lost those shared with human luteinizing hormone (hLH). β -hCG has been investigated as a contraceptive vaccine to aid population control, since antibodies to β -hCG neutralize the hormone and

prevent successful pregnancy. However, the production of antibodies to hLH is undesirable since this hormone is present all the time, not just during pregnancy as is essentially the case for hCG (cf. figure 17.21). The mutant epitopes (orange) avoid this unwanted LH cross-reactivity. The C-terminal end of the construct can be extended to encode suitable T-helper carrier epitopes. For simplicity, the T-cell recognition of MHC-peptide has been ignored. (b) FACS analysis of cells transfected to express β -hCG on their surface, showing a mutant which retains the β -hCG-specific epitope but has lost the cross-reactive epitopes characteristic of the wild-type protein.

Likewise, the isolated 19kDa C-terminal fragment of the merozoite-specific protein MSP-1 confers antibody-mediated protection on monkeys to challenge with the blood stage of the malaria parasite.

CURRENT VACCINES

The established vaccines in current use and the schedules for their administration are set out in tables 14.3 and 14.4.

Because of the pyrogenic reactions and worries about possible hypersensitivity responses to the whole-cell component of the conventional pertussis vaccine, a new generation of vaccines containing one or more purified components of *Bordetella pertussis*, and therefore termed 'acellular' vaccines, has been licensed in the USA. The combination of diphtheria and tetanus toxoids with acellular pertussis (DTP) is recommended for the later fourth and fifth 'shots' and for children at increased risk for seizures. Children under 2 years of age make inadequate responses to the T-independent *H. influenzae* capsular polysaccharide, so they are now routinely immunized with the antigen conjugated with tetanus or diphtheria toxoids.

A resurgence of **measles** outbreaks in recent years has prompted recommendations for the reimmunization of children, at the age of school entry or at the change to middle school, with measles. The considerable morbidity and mortality associated with hepatitis B infection, its complex epidemiology and the diffi-

culty in identifying high-risk individuals have led to recommendations for vaccination in the 6–18-month age group.

Maternally derived IgG antibody can inhibit *de novo* immune responses through feedback control (cf. p. 201). Preliminary results suggest that infants of 4–6 months can be seroconverted by inhaled aerosol measles vaccine which presumably evades the maternal antibody; this will have singular relevance in endemic measles areas, where almost split-second timing is required with conventional immunization as passively acquired antibody wanes. Naked DNA vaccines may also have a role to play here.

Tuberculosis is the largest cause of death in the world from a single infectious disease and, while it remains a truly major problem in developing countries, cases have also increased by around one-third in Western countries. There is an alarmingly heightened susceptibility to TB of individuals with HIV, and worldwide multidrug-resistant strains are appearing. Thus, although BCG has been in use for 70 years and is reasonably efficacious and safe in healthy non-T-cell-deficient subjects, there is certainly a vital need for the development of new drugs and vaccines. It remains to be seen whether the disappointing degree of protection against TB and *Mycobacterium leprae* found with BCG in field trials in certain parts of the globe is due to the use of high doses, deficient strains or subversion of the response to group i cross-reacting antigens by suppressive mycobacterial species in the local environment.

Table 14.3. Current and experimental vaccines.

Type	Established	Experimental
BACTERIAL VACCINES		
Live attenuated	<i>Mycobacterium tuberculosis</i> (BCG)	● <i>Vibrio cholerae</i> ● <i>Salmonella typhi</i> (Ty21a: Vi⁺ mutant) ● <i>S. typhi</i> (aroA: aromatic pathway mutant)
Inactivated	<i>V. cholerae</i> <i>B. pertussis</i> <i>S. typhi</i>	● <i>V. cholerae</i> + toxin B subunit ● <i>M. leprae</i>
Subunit	<i>Haemophilus influenzae</i> <i>Neisseria meningitidis</i> <i>Streptococcus pneumoniae</i>	● <i>S. typhi</i> (capsular Vi⁺ carrier) ● <i>H. influenzae</i> (dip/tet toxoid) ● <i>M. tuberculosis</i> (naked DNA)
Toxoid	Tetanus, diphtheria	
VIRAL VACCINES		
Live attenuated	Vaccinia Measles Mumps Rubella ● Polio (Sabin) ● Varicella zoster ● Adeno Yellow fever	● Cytomegalovirus ● Hepatitis A ● Influenza ● Dengue ● Rota ● Parainfluenza ● Japanese encephalitis
Inactivated	Polio (Salk) Influenza Rabies Japanese encephalitis Hepatitis A	
Subunit	Hepatitis B, influenza	HIV, influenza, hepatitis B/C, herpes, cytomegalovirus, rabies (all naked DNA in experimental animals)

● Given orally

EXPERIMENTAL VACCINES IN DEVELOPMENT

Malaria

Don't sneer at low technology. The major advance in malaria control has been the finding that the impregnation of bed nets with the insecticide pyrethroid reduces *Plasmodium falciparum* deaths by 40%. However, with the emergence of drug-resistant strains of mosquito, vaccines must be developed. The goal is achievable since, although children are very susceptible, adults resident in highly endemic areas acquire a protective but nonsterilizing immunity probably mediated by antibodies specific for the blood stages.

Antigen variation poses a big problem for vaccine development and a number of investigators in the

Table 14.4. Current vaccination practice.

VACCINE		ADMINISTRATION		
		UK	USA	OTHER COUNTRIES
CHILDREN				
Triple (DTP) vaccine: diphtheria, tetanus, pertussis	Primary	2–6 mo (3x/4 weekly)		Japan: 2 yr
	Boost DT	3–5yr	15 mo/4 yr DT every 10 yr	
Polio: live	Primary	Concomitant with DTP		
	Boost	4/6 yr	15 mo, 4 yr high-risk adult	
		Immunocompromised		
killed				
MMR vaccine: measles, mumps rubella	Primary	12–15 mo		Africa: 6 mo
	Boost	4–5yr 10–14 yr seronegative girls selectively with rubella		
BCG (TB, leprosy)		10–14 yr	high risk only	Tropics: at birth
Haemophilus		Concomitant with DTP		
Varicella		Neonates at risk, immunocompromised		
ELDERLY				
Pneumococcal polysaccharide serotypes		Aged & high risk		
Influenza		Aged & high risk		
SPECIAL GROUPS				
Hepatitis B		Travellers, high risk groups		
Hepatitis A Meningitis (A+C)		Travellers to endemic areas		
Yellow fever Typhoid		Travellers to endemic areas		Tropics: infants Yellow fever: boost residents and frequent visitors every 10 yr
Rabies		Prophylactically in high risk groups Post-exposure to contacts in endemic areas		

malaria field have turned their attention to the invariant antigens of the **sporozoite**, which is the form with which the host is first infected; this rapidly reaches the liver to emerge later as merozoites which infect the red cells (figure 14.14). Because the sporozoite only takes 30 minutes to reach the liver, the antibody has to act fast, so that inactivation is limited by diffusion events and hence dependent on the concentration of antibody molecules. The sporozoite has a characteristic antigen with multiple tetrapeptide repeats. *Plasmodium falciparum*, for example, has 37 repeats of NANP. Field trials of vaccines with polymers of NANP and similar tetrapeptides from other species have not so far been spectacularly successful, but building in powerful T-cell help in the form of the promiscuous 378–398

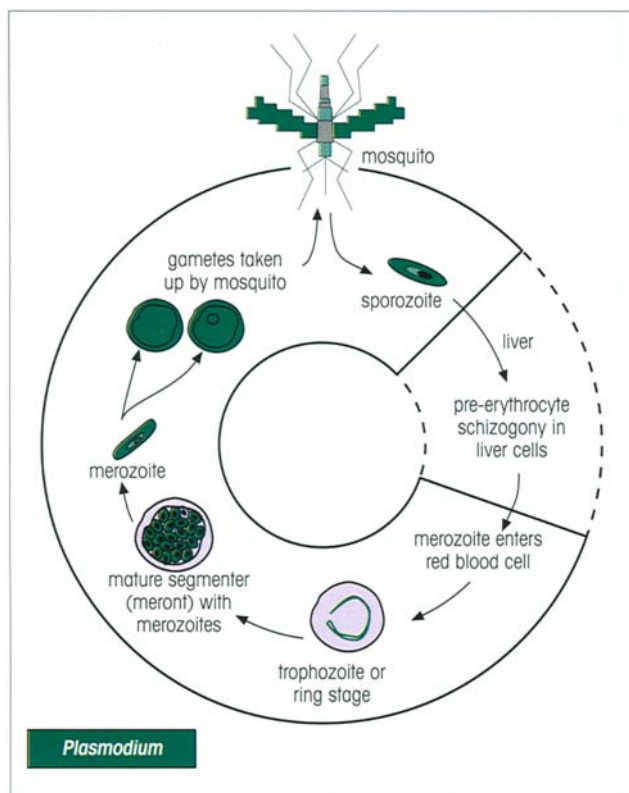


Figure 14.14. The life cycle of the malaria parasite.

epitope (see p. 294) or a mycobacterial heat-shock protein hsp70 carrier (which obviates the need for adjuvant) should improve efficacy. Mice immunized with radiation-attenuated sporozoites or circumsporozoite protein linked to attenuated *Salmonella* were resistant to live challenge with sporozoites, which should be useful for nonimmune travelers visiting areas where malaria is endemic. Immunity is mediated by noncytotoxic CD8 cells which target infected hepatocytes through the production of $\text{IFN}\gamma$, $\text{NO}\cdot$ and IL-12 ; the latter cosily initiates a positive feedback loop by stimulating $\text{IFN}\gamma$ secretion by NK cells.

Many laboratories have been targeting the various antigens associated with the **blood stages**, and much publicity has been given to trials using the vaccine SPf66, which consists of three peptide epitopes from three blood stage proteins intercalated with NANP sequences. Some clear protection was originally obtained in large-scale trials in South America, but two further trials in other areas were unfortunately ineffective. The C-terminal fragment of the merozoite-specific protein, MSP-1₁₉, mentioned earlier, has considerable promise. One of the products of the infected erythrocytes is a glycopospholipid with features of the insulin second messenger. This may be responsible

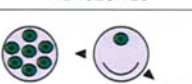
STAGE	VACCINE	STRATEGY
 SPOROZOITES	CIRCUMSPOROZOITE (CS) PROTEIN OR PEPTIDES	Ab blocks infection of liver
 LIVER STAGE	CS PROTEIN OR PEPTIDES	T-cell mediated immunity destroys liver stage
 MEROZOITES	MEROZOITE ANTIGENS	Block infection of red cells
 ASEXUAL ERYTHROCYTE STAGE	ASEXUAL STAGE RBC	Destroy infected red cells
 TOXIC ANTIGEN	TOXIC PRODUCT	Ab neutralizes toxin
 GAMETOCYTES	SEXUAL STAGES	Ab blocks transmission
 GAMETES		
 MOSQUITO	MOSQUITO TRYPSIN	

Figure 14.15. Strategies for a malaria vaccine. A vaccine incorporating a yeast-derived polypeptide representing the zygote-specific antigen Pfs25, which can induce transmission-blocking antibodies, looks like the next candidate for big field trials of efficacy. If antibodies to the mosquito midgut trypsin are taken into the blood meal with the gametocytes, they block the trypsin-mediated activation of the parasite pro-chitinase enzyme required for penetration of the chitin layer lining the midgut by the ookinates (differentiated from the gametocyte stage).

for many of the secondary features of malaria by acting as an excessive stimulator of TNF production. A strategy based on using the molecule as a vaccine would resemble the well-established approach to tetanus and diphtheria, where the toxin rather than the organism is attacked. Other groups are focusing on antigens which elicit antibodies able to **block transmission** (see legend to figure 14.15); these would be altruistic vaccines in that they only help the next guy down the line and finally the community as a whole.

A brief summary of the many different vaccine strategies being deployed to counter this complex parasite is presented in figure 14.15, and there may be a consensus that, ultimately, vaccines will contain antigens from all stages probably presented as a mixture of the naked DNA genes which encode them. Perhaps the time has come to introduce a radically new attack on the problem, side-stepping the relatively small-scale studies currently in progress. A gigantic shotgun approach to the identification of protective sporozoite vaccine candidates requiring a hefty input of governmental investment has been proposed. The strategy envisages the following stages: (i) identify putative open reading frames (ORFs) from the genomic sequence of the parasite, (ii) immunize mice with DNA vaccine plasmids derived from each ORF, (iii) use the antisera to screen hepatocytes infected with irradiated sporozoites, (iv) select those plasmids shown by the selection process to encode antigens expressed in the hepatocytes, (v) predict degenerate HLA superfamily binding T-cell epitopes, (vi) validate them as targets of CD8 responses in volunteers immunized with irradiated sporozoites, (vii) string the resulting T-cell epitopes together in a single vaccine plasmid. Phew! This 'big picture' post human-genome style may well be the blueprint for the future.

Schistosomiasis

The morbidity in this chronic and debilitating disease is related to the remarkable fecundity of the female worm which lays hundreds of eggs every day. These are deposited in numerous mucous membranes and tissues, and the granulomas which form around them (cf. figure 16.28) lead to the development of severe fibrotic and often irreversible lesions. Specific IgE can produce worm damage through antibody-dependent cellular cytotoxicity (ADCC) mechanisms (cf. p. 32 and figure 13.22) involving eosinophils and possibly mononuclear phagocytes as effector cells, and may be regarded as a factor in recovery from infection together with Th1 CMI, schistosomes being susceptible to the lethal effects of NO $\cdot$ produced by activated macrophages. IgE, G and A antibodies correlate with resistance to reinfection in drug-cured patients, while IgA appears to control fecundity as shown by passive transfer experiments. Our abilities to stimulate IgE and, to a lesser extent, IgA antibodies at will are, to put it mildly, still somewhat underdeveloped, but it is encouraging to note that a monoclonal antibody to the glutathione-S-transferase of *S. mansoni* inhibits the enzyme, protects against challenge and dramatically reduces the laying and viability of eggs.

Another protective monoclonal to a different epitope has no effect on enzymic activity or on the production and viability of eggs, but does reduce the worm burden. The recombinant enzyme, possibly in a cocktail with other schistosome antigens and cercarial proteases, could provide a promising future vaccine.

Cholera

In the case of **cholera**, an oral vaccine which combines the B subunit of cholera toxin with killed vibrios has been reported in a big field trial in Bangladesh to stimulate excellent gut mucosal antibody formation, the response being said to equal that seen after clinical cholera. The vaccine also afforded cross-protection against the enterotoxin of *E. coli*, responsible for travelers' diarrhea.

Tuberculosis

Extensive field trials of the nonvirulent *Mycobacterium vaccae* are underway using the killed organisms which generate strong Th1 responses. To obviate the potential adverse effects of live BCG vaccine in immunodeficient subjects (e.g. HIV-positive), auxotrophic strains which fail to grow in the absence of essential amino acids, such as methionine and leucine, were created by insertional mutagenesis. The strains died out within 16–32 weeks in mice and severe combined immunodeficient (SCID) mice survived for at least 230 days compared with 8 weeks for a conventional BCG vaccine. These auxotrophic strains gave excellent protection against challenge with virulent bacilli and so offer a potentially safe vaccine against TB for populations at risk for HIV.

At the subunit level, as might now be anticipated, naked DNA vaccines for heat-shock and other common mycobacterial antigens are being developed and seem to be highly effective, in mice at least.

ADJUVANTS

For practical and economic reasons, prophylactic immunization should involve the minimum number of injections and the least amount of antigen. We have referred to the undoubted advantages of replicating attenuated organisms in this respect, but nonliving organisms, and especially purified products, frequently require an adjuvant which, by definition, is a substance incorporated into or injected simultaneously with antigen which potentiates the immune response (Latin *adjuvare*—to help). It is interesting that bacterial structures provide the major source of immunoadjuvants,

presumably because they provide danger signals of infection, and it is no accident that the 'piggy-back' incorporation of the gene encoding an immunogen into attenuated *Salmonella* and BCG can be so effective. In a sense, the basis of adjuvant activity is often the recognition of these signals by phylogenetically ancient receptors on accessory cells. The mode of action of adjuvants may be considered under several headings.

Depot effects

Free antigen usually disperses rapidly from the local tissues draining the injection site, and an important function of the so-called repository adjuvants is to counteract this by providing a long-lived reservoir of antigen, either at an extracellular location or within macrophages. The most common adjuvants of this type used in humans are **aluminum compounds** (phosphate and hydroxide). Empirically, it has long been realized that hydrophobic substances tend to augment immune responses, and Freund's incomplete adjuvant, in which the antigen is incorporated in the aqueous phase of a stabilized water-in-paraffin oil emulsion, usually produces higher and far more sustained antibody levels with a broadening of the response to include more of the epitopes in the antigen preparation. However, because of the lifelong persistence of oil in the tissues and the occasional production of sterile abscesses, and the unpalatable fact that paraffin oil produces tumors in mice, the replacement of incomplete Freund's with different types of oil, such as squalene or biodegradable peanut oil, has been considered. It is reassuring that the recent analysis of large-scale trials of vaccines containing emulsified mineral oils, performed 30–40 years ago, has shown that such procedures do not increase the incidence of neoplasms or autoimmune disease.

Activation of antigen-presenting cells

Under the influence of the repository adjuvants, macrophages form granulomas which provide sites for interaction with antibody-forming cells. The maintenance by the depot of consistent antigen concentrations ensures that, as antigen-sensitive cells divide within the granuloma, their progeny are highly likely to be further stimulated by antigen. Virtually all adjuvants deliver antigens to antigen-presenting cells and stimulate them by improving immunogenicity through an increase in the concentration of processed antigen on their surface and the efficiency of its presentation to lymphocytes, by the provision of accessory costimulatory signals to direct lymphocytes towards

an immune response rather than tolerance, and by the secretion of soluble stimulatory cytokines which influence the proliferation of lymphocytes. For example, quite apart from the ability of mycobacterial hsp70 to act as a powerful carrier, its innate adjuvant activity (figure 14.7) is mediated by upregulating expression of the proinflammatory cytokines, TNF, IL-1 β and IL-6, through a CD14/NF κ B pathway. One of the most potent stimulators of antigen-presenting cells is lipid A from Gram-negative bacterial lipopolysaccharide (LPS), but it has many side-effects and interest has shifted to its derivative, monophosphoryl lipid A (MLA), which is less toxic. The **Ribi adjuvant**, perhaps one of the most commonly used formulations in experimental work (and hopefully in humans), is a water-in-oil emulsion incorporating MLA and an MDP derivative (see below).

Effects on lymphocytes

In mice, alum tends to stimulate helper cells of the Th2 family, whereas complete Freund's adjuvant favors the Th1 subset. It will be recalled that complete Freund's is made from the incomplete adjuvant by addition of killed mycobacteria (cf. p. 192), but the immunopathological effects of the mycobacterial component in complete Freund's are so striking that its use in humans is not normally countenanced; fortunately, the active component has been identified as the water-soluble **muramyl dipeptide** (MDP; *N*-acetylmuramyl-L-alanyl-D-isoglutamine) and a number of acceptable derivatives are now available. Hydrophilic MDP analogs with aqueous antigen preferentially stimulate antibody responses, but if administered in a hydrophobic microenvironment, such as mineral oil, or incorporated into liposomes, CMI is the major outcome.

The role of modulatory cytokines in these early immunologic interactions is important, and several experimental approaches are exploring the effect of cytokines administered simultaneously with antigen. In one study, a construct of the macrophage stimulator granulocyte-macrophage colony-stimulating factor (GM-CSF), linked to a monoclonal immunoglobulin, successfully induced anti-idiotypic responses relevant to the treatment of chronic lymphocytic leukemia. IL-2 has an adjuvant activity in unresponsive leprosy patients and a single injection in dialysis patients effectively induced their seroconversion to hepatitis B surface antigen. The potential of IL-12 to encourage Th1 responses is also being actively pursued.

The need for adjuvants which stimulate mucosal immunity is being met by exploiting the ability of *E. coli*

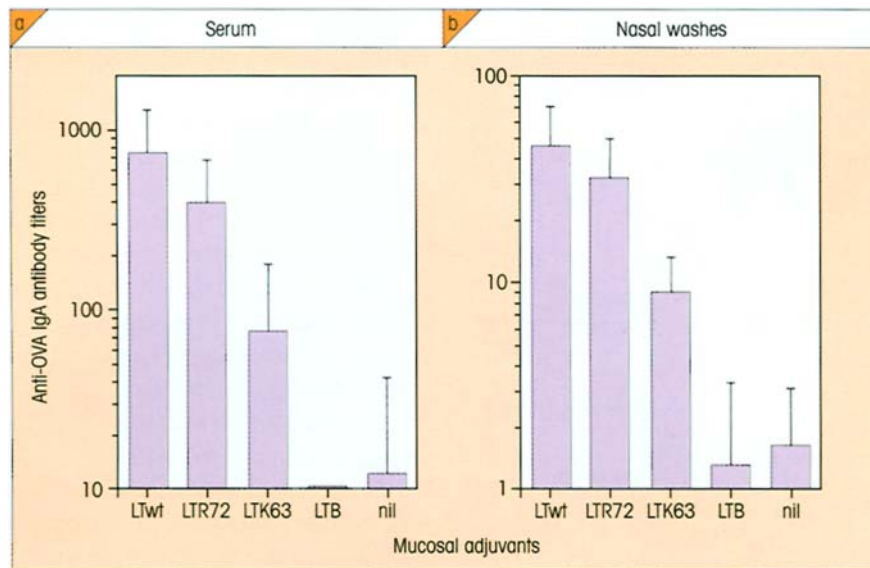


Figure 14.16. Mucosal adjuvanticity of mutants of *E. coli* heat-labile enterotoxin (LT) with reduced toxicity. Ovalbumin-specific IgA antibody titers (mean+SD) in mice immunized intranasally three times with ovalbumin alone or in combination with wild-type LT (LTwt), an A → R mutant of the A chain at position 72 (LTR72) retaining only 0.6% of the toxic ADP-ribosyltransferase activity, a non-toxic S → K mutant of the A chain at position 63 (LTK63), and the

recombinant B chain (LTB) which targets the monosialoganglioside (GMI) receptor-binding site. LTR72 has comparable adjuvanticity to LTwt but only very low toxicity (LTwt is responsible for travelers' diarrhea). Even higher titers of antibody are recovered from lung washes and lesser amounts from vaginal washes. (Data from Giuliani M.M. *et al.* (1998) *Journal of Experimental Medicine* 187, 1123, reproduced with permission.)

heat-labile enterotoxin (LT) and cholera toxin to target intestinal cells. LT is very immunogenic when given by the oral route and a nontoxic mutant of comparable potency as an adjuvant (figure 14.16) has been developed for ultimate use in humans. Coupling a protein antigen to cholera toxin B subunit targets the vaccine to the epithelial cells of the intestinal tract and usually produces good IgG and IgA antibodies which appear also in the saliva, tears and milk, indicating that the antigen-specific IgA precursor cells become disseminated throughout the MALT system. Whether the immune response to the cholera precludes its use as a carrier for immunization with further antigens remains an open question. Speaking of cholera toxin, it was reported that a soluble antigen, such as diphtheria toxoid, applied to the skin of mice together with cholera toxin produced persistent high levels of IgG antibodies, a not entirely predictable outcome.

Newer approaches to the presentation of antigen

Recent interest has centered on the use of small lipid membrane vesicles (**liposomes**) as agents for the presentation of antigen to the immune system. It may be that the liposome acts as a storage vacuole within the macrophage or perhaps fuses with the macrophage membrane to provide a suitably immunogenic com-

plex. The differing pathways for processing peptides within antigen-presenting cells can be turned to advantage by encapsulating antigen in acid-resistant liposomes so that they can only enter the MHC class I route and stimulate CD8 T-cells. Antigens within acid-sensitive liposomes become associated with both class I and class II molecules. Proteins anchored in the lipid membrane by hydrophobic means give augmented CMI. Short synthetic peptides coupled covalently to monophosphoryl lipid A or tripalmitoyl-S-glycerylcysteinyl-seryl-serine (P3CSS) have high priming efficiency. One can readily envisage the possibility of a single-shot liposome vaccine with multiple potentialities which incorporate several antigens, different adjuvants and specialized targeting molecules (figure 14.17).

Another innovation is the **Iscom** (immunostimulating complex), a hydrophobic matrix of the surfactant saponin, with antigen, cholesterol and phosphatidylcholine. Antigens with a transmembrane hydrophobic region, such as surface molecules of lipid-containing viruses, are powerfully immunogenic in this vehicle and may engender cytotoxic T-cell responses. Iscoms are extremely resistant to acid and bile salts and are immunogenic by the oral route, producing systemic immunity and good local secretory IgA. Intranasal exposure established protective immunity to influenza

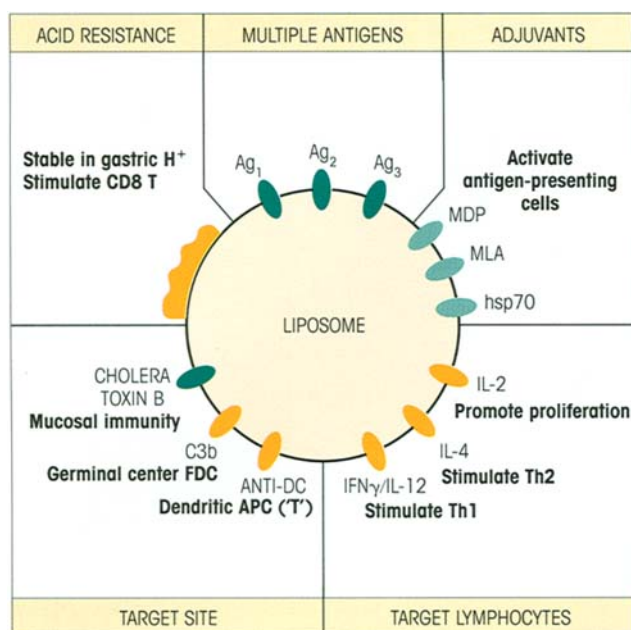


Figure 14.17. The 'do-it-all-in-one' omnipotent liposome particle illustrating some possible ways to build immunogenic flexibility into a single-shot liposome vaccine. The interior of the liposome may also contain depots of certain components. MDP, muramyl dipeptide; MLA, monophosphoryl lipid A; hsp70, mycobacterial heat-shock protein 70.

in mice. Saponin itself, initially purified from the bark of the tree *Quillaja saponaria*, is too toxic for human use. A less toxic derivative, Quil A, is widely used for veterinary vaccines, but an even less toxic derivative, QS21, is being evaluated for safety and efficacy in clinical trials. Another group of surfactants, the non-ionic block copolymers having a hydrophobic poly-

oxypropylene core linked to hydrophilic polymers of polyoxyethylene, are likely to be acceptable for use in humans.

It may be useful to focus on the notion floated earlier that polymeric antigens tend to be more immunogenic, and to note a novel form of solid matrix of the Cowan strain of *S. aureus* which can bind several molecules of monoclonal antibody, which can, in turn, immobilize several molecules of antigen. Using a variety of monoclonals, one can purify onto their binding sites appropriate antigens from a mixture to give a multivalent subunit vaccine. One could also incorporate the *E. coli* enterotoxin mutants mentioned above to give good mucosal immunity.

With respect to the practicability and convenience of administration of vaccines, we should not ignore the use of the **biolistics gun** (cf. p. 141) to introduce gold microspheres coated with plasmids of the gene encoding the vaccine antigen or antigens (cf. p. 290).

There have been some very important advances in the design of controlled-release systems for antigen delivery *in vivo*. Polymers of polylactic–polyglycolic esters are nontoxic biodegradable vehicles which can be prepared in a mixture of different formulations to provide **slow release** of an antigen (or any active drug or hormone) for periods up to several months.

There are exciting times ahead for vaccine development, but it should always be borne in mind that no matter how successful these clever strategies prove to be in experimental animals, the acid test which destroys so many promising approaches is the long-term effect in the human.

SUMMARY

Passively acquired immunity

- Passive immunity can be acquired by maternal antibodies or from homologous pooled γ -globulin.
- Horse antisera are more restricted because of the danger of serum sickness.
- Antibodies are being constructed to order using recombinant DNA technology and can be produced in bulk in plants.

Vaccination

- Active immunization provides a protective state through contact with a harmless form of the disease organism.
- A good vaccine should be based on antigens which are

easily available, cheap, stable under extreme climatic conditions and nonpathogenic.

Killed organisms as vaccines

- Killed bacteria and viruses have been widely used.

Live attenuated organisms

- The advantages are: replication gives a bigger dose, the immune response is produced at the site of the natural infection.
- Attenuated vaccinia or polio can provide a 'piggy-back' carrier for genes from other organisms which are difficult to attenuate.
- BCG is a good vehicle for antigens requiring CD4 T-cell

(continued)